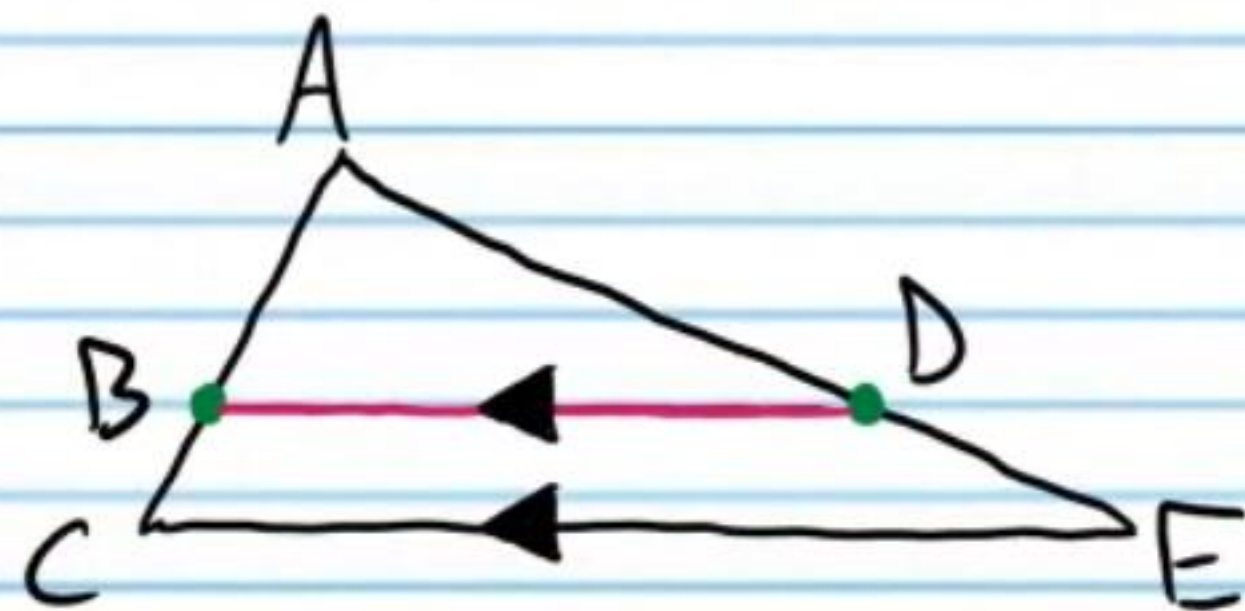


Proportionality Theorems, Transformations, and Perimeter and Area of Similar Polygons

Proportionality Theorems

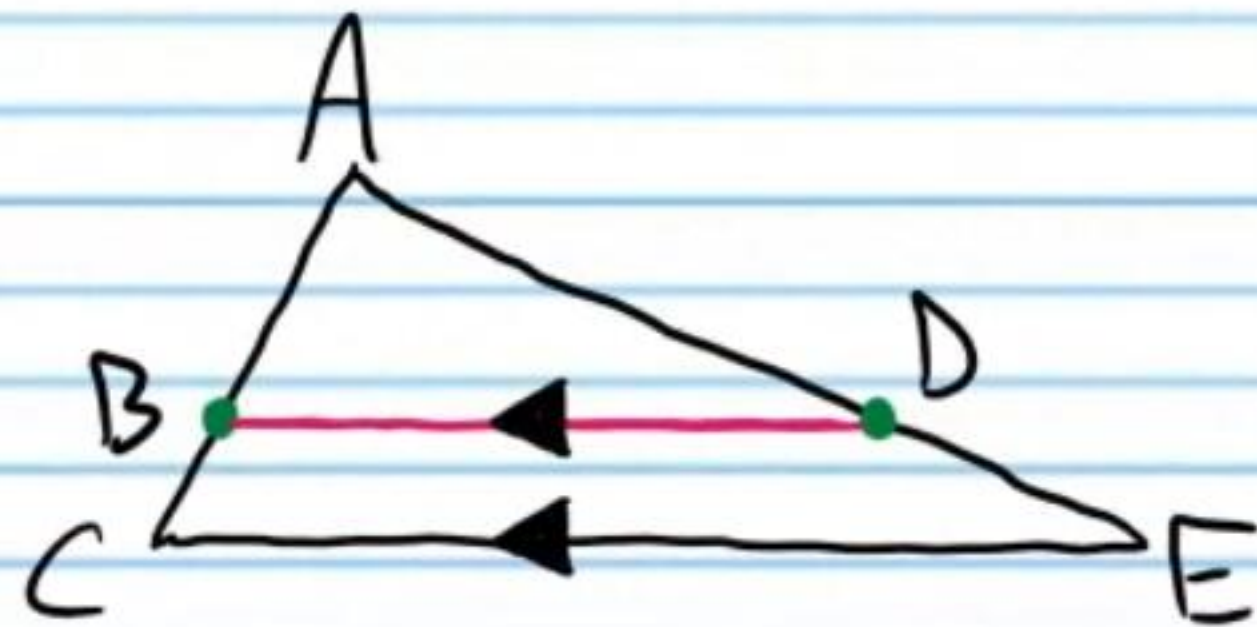
Triangle Proportionality Theorem: If a line parallel to one side of a triangle intersects the other two sides, then it divides the two sides proportionally.

If $\overline{BD} \parallel \overline{CE}$,
then $\frac{AB}{BC} = \frac{AD}{DE}$.

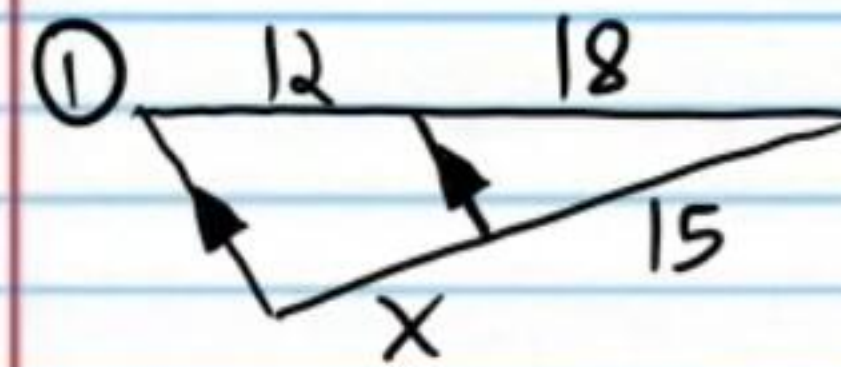


Triangle Proportionality Converse Theorem: If a line divides two sides of a triangle proportionally, then it is parallel to the third side.

If $\frac{AB}{BC} = \frac{AD}{DE}$,
then $\overline{BD} \parallel \overline{CE}$.



Find the unknown values.

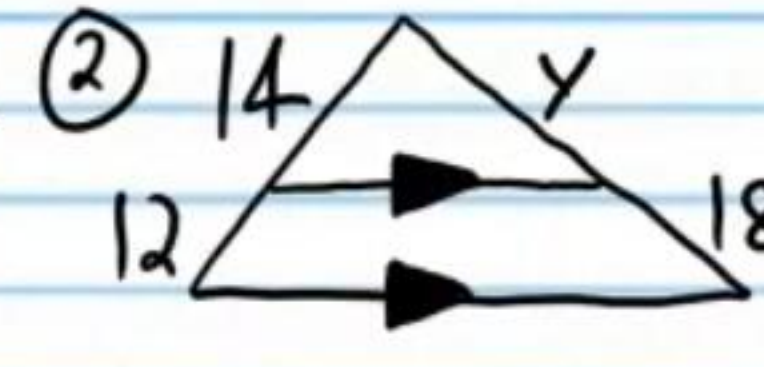


$$\frac{18}{12} = \frac{15}{x}$$

$$\frac{3}{2} = \frac{15}{x}$$

$$3x = 30$$

$$\boxed{x = 10}$$

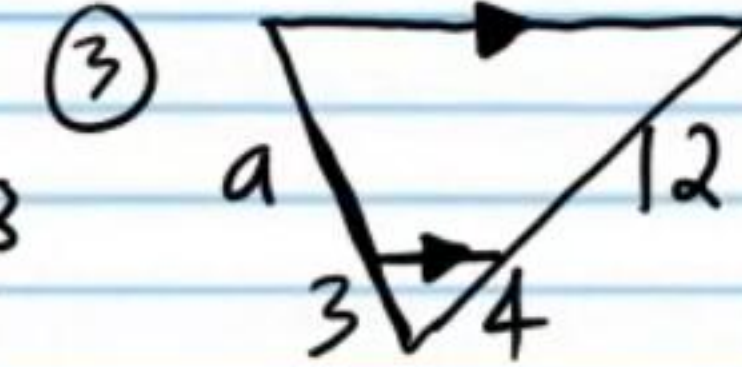


$$\frac{14}{12} = \frac{y}{18}$$

$$\frac{7}{6} = \frac{y}{18}$$

$$6y = 126$$

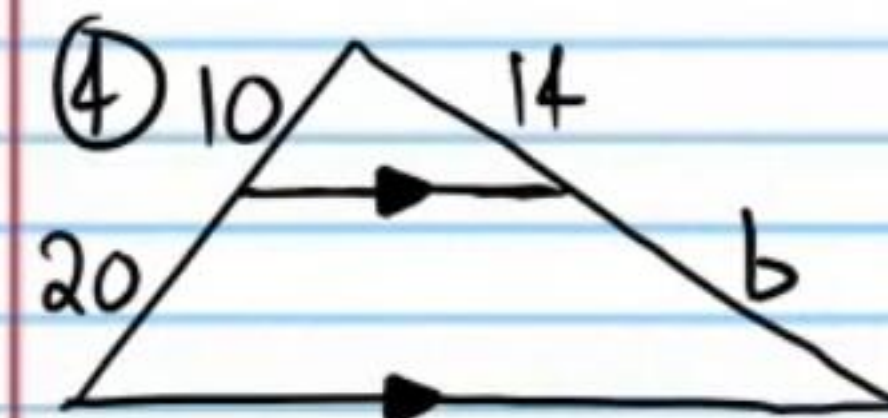
$$\boxed{y = 21}$$



$$\frac{a}{3} = \frac{12}{4}$$

$$\frac{a}{3} = \frac{3}{1}$$

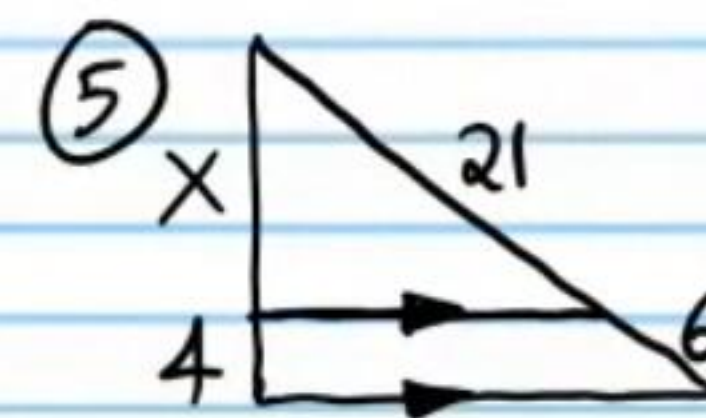
$$\boxed{a = 9}$$



$$\frac{10}{20} = \frac{14}{b}$$

$$\frac{1}{2} = \frac{14}{b}$$

$$\boxed{b = 28}$$

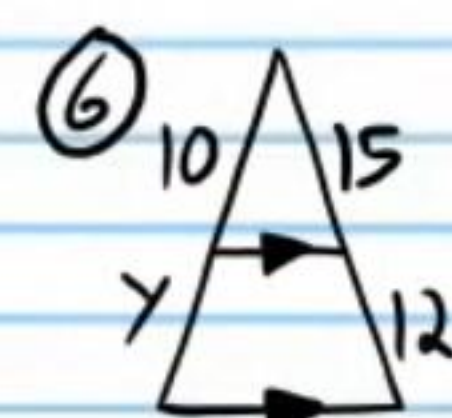


$$\frac{x}{4} = \frac{21}{6}$$

$$\frac{x}{4} = \frac{7}{2}$$

$$2x = 28$$

$$\boxed{x = 14}$$



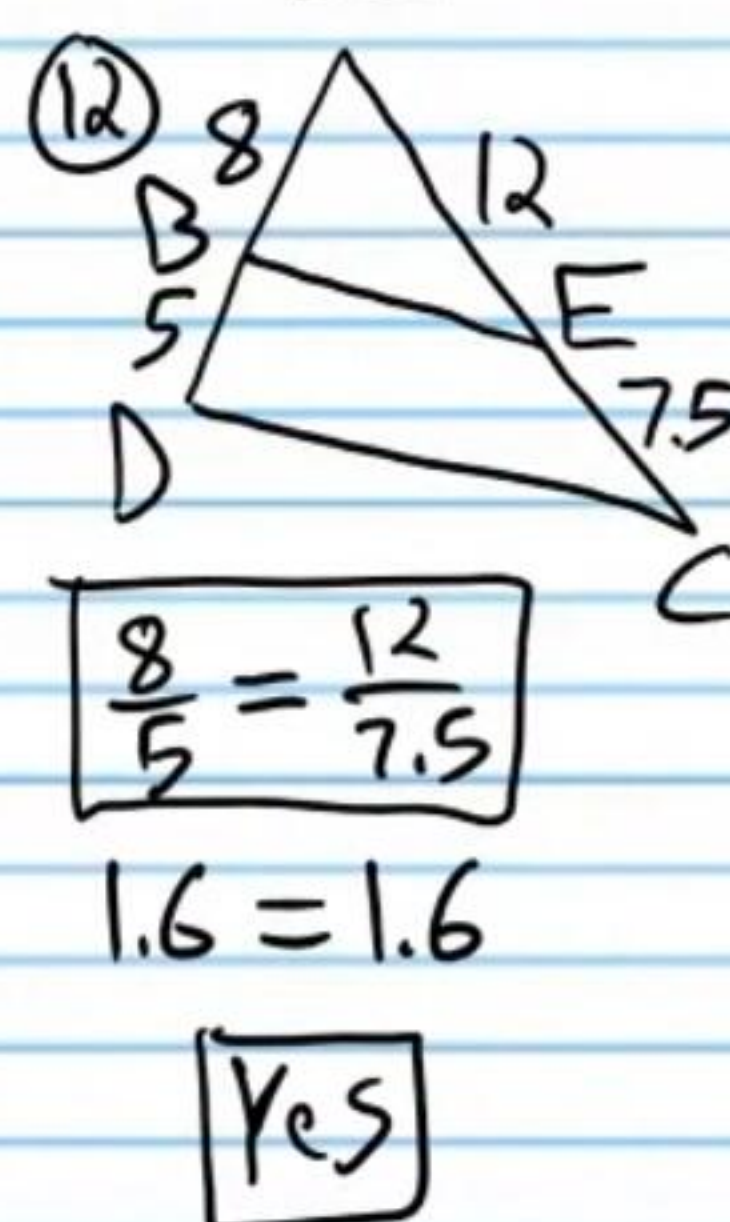
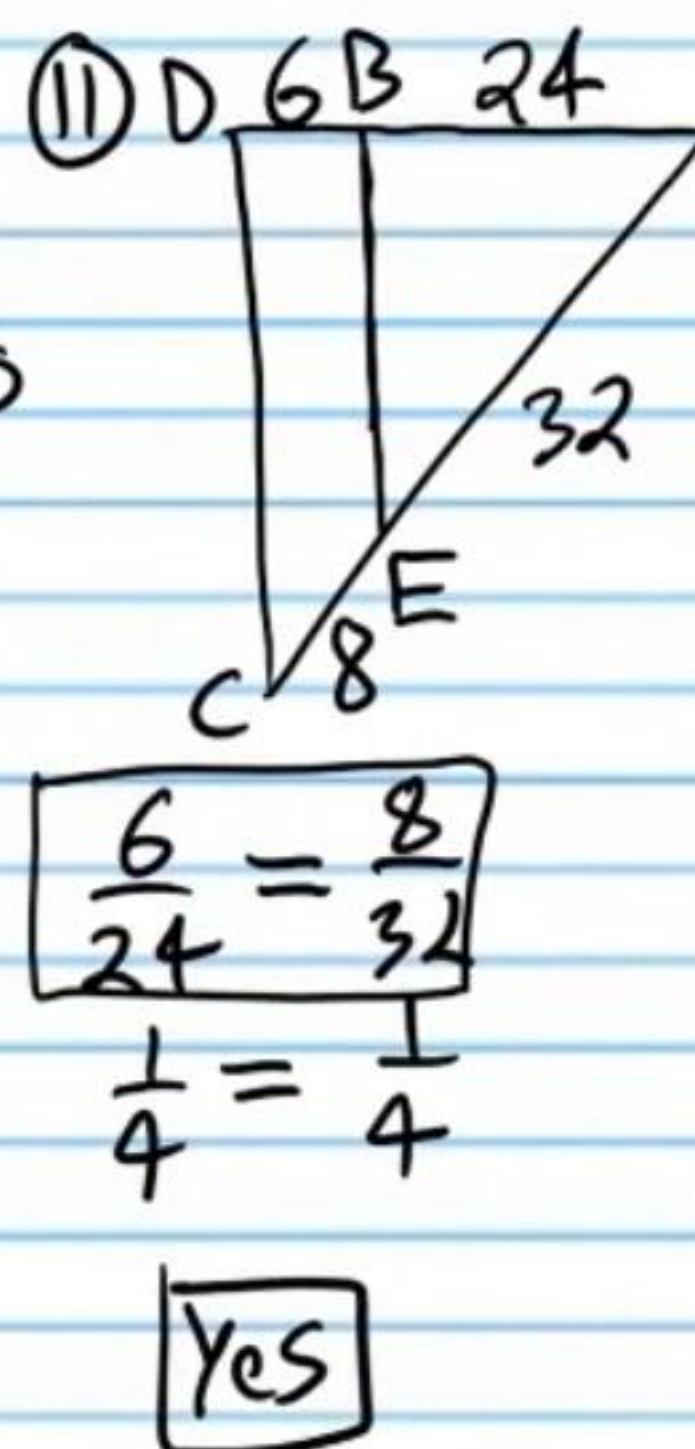
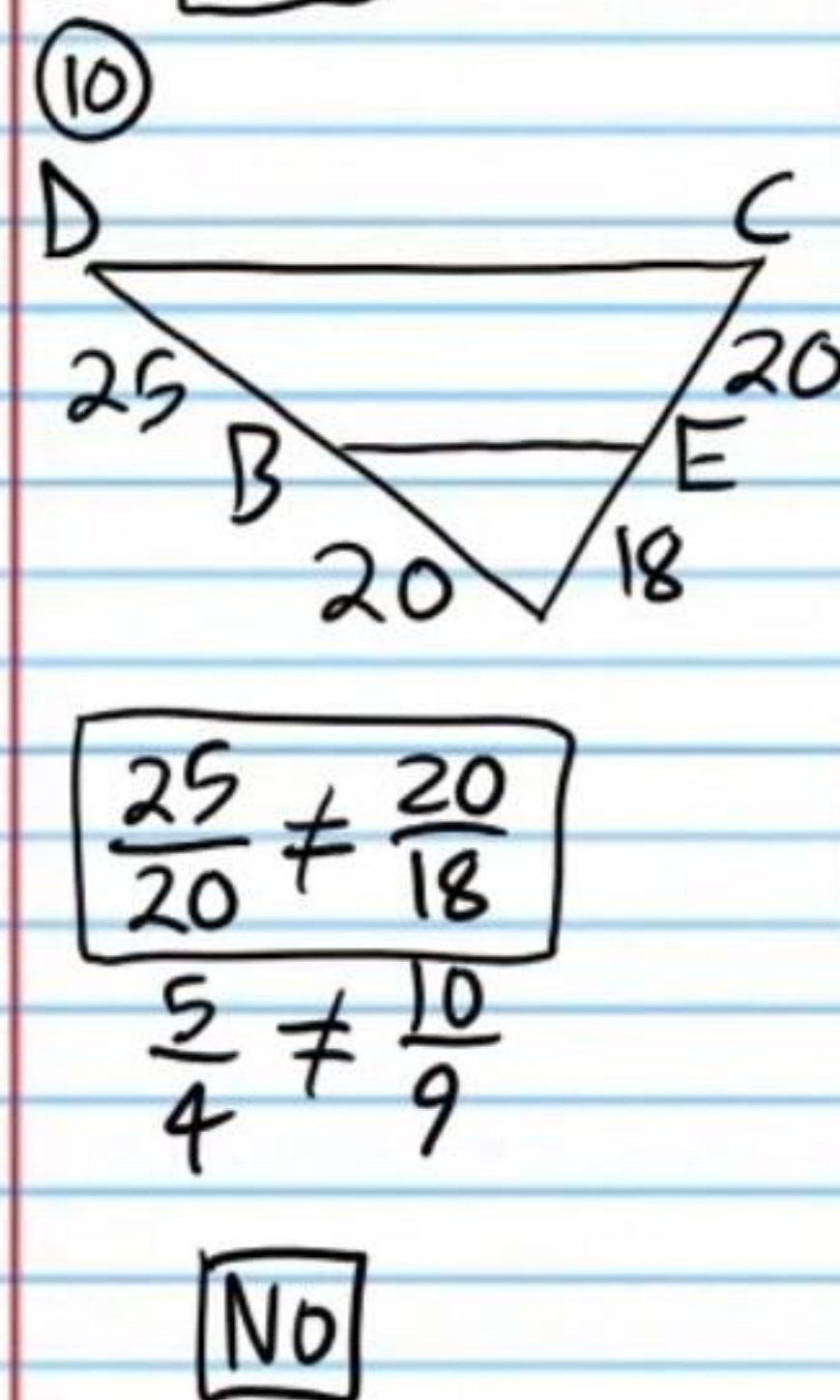
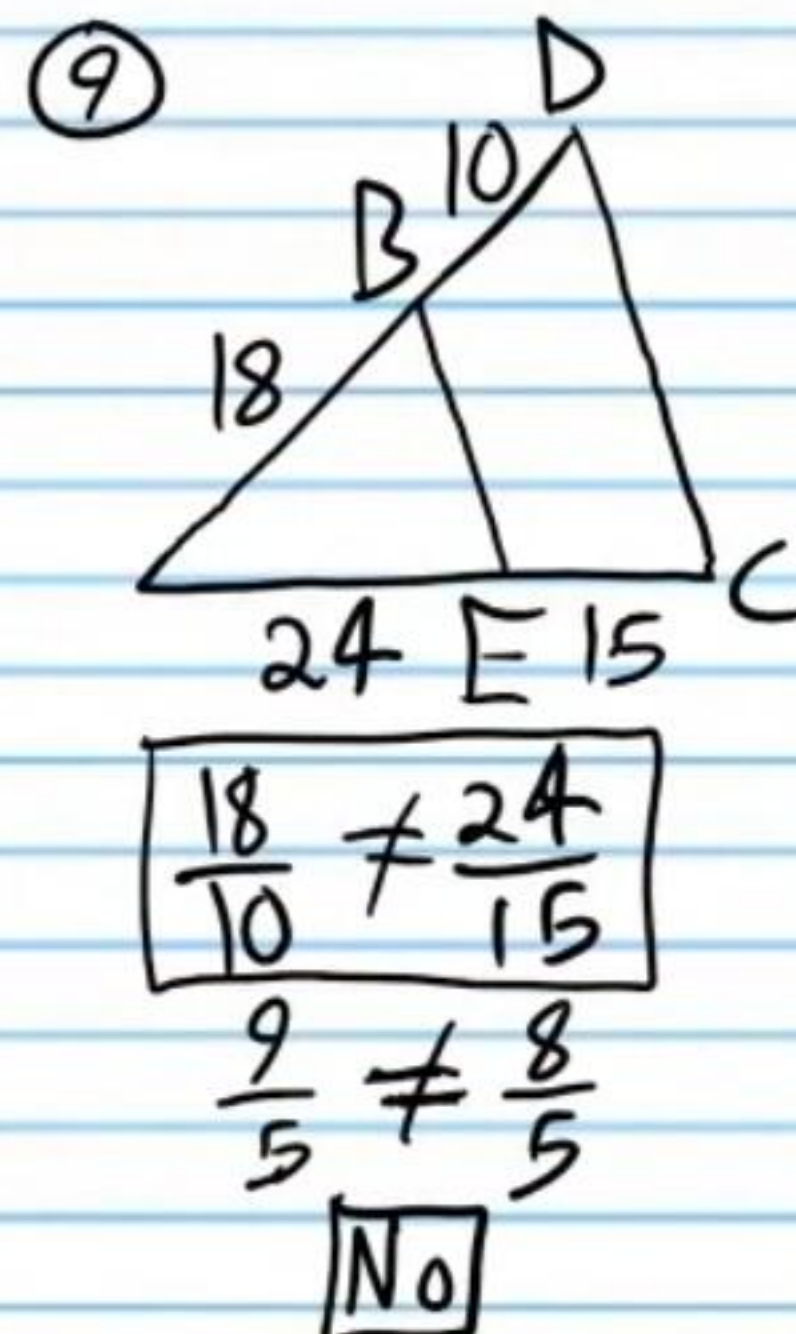
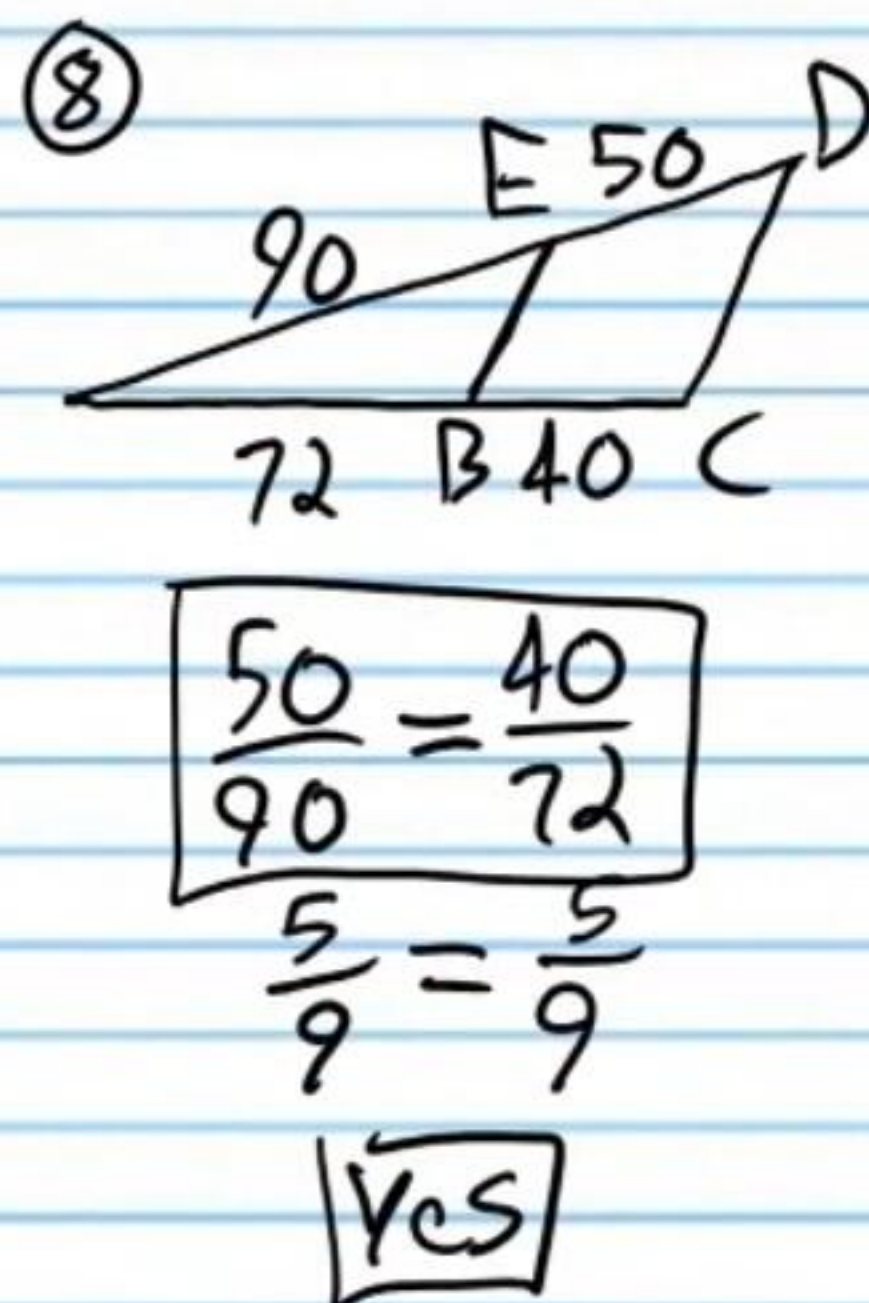
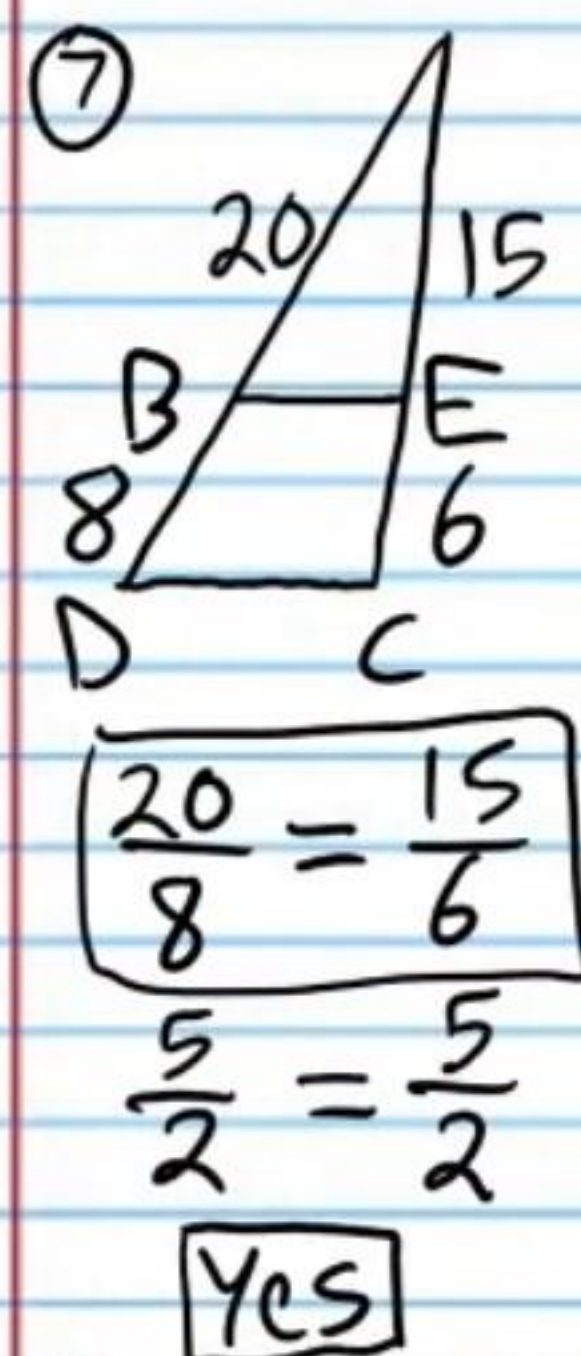
$$\frac{10}{y} = \frac{15}{12}$$

$$\frac{10}{y} = \frac{5}{4}$$

$$40 = 5y$$

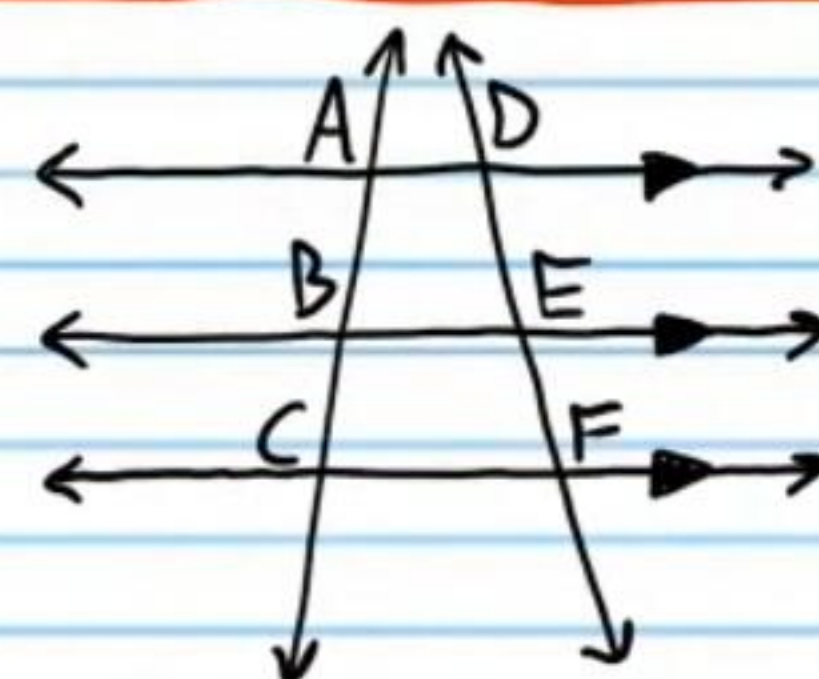
$$\boxed{8 = y}$$

Is $\overline{BE} \parallel \overline{DC}$? Show the proportions that prove your answers.

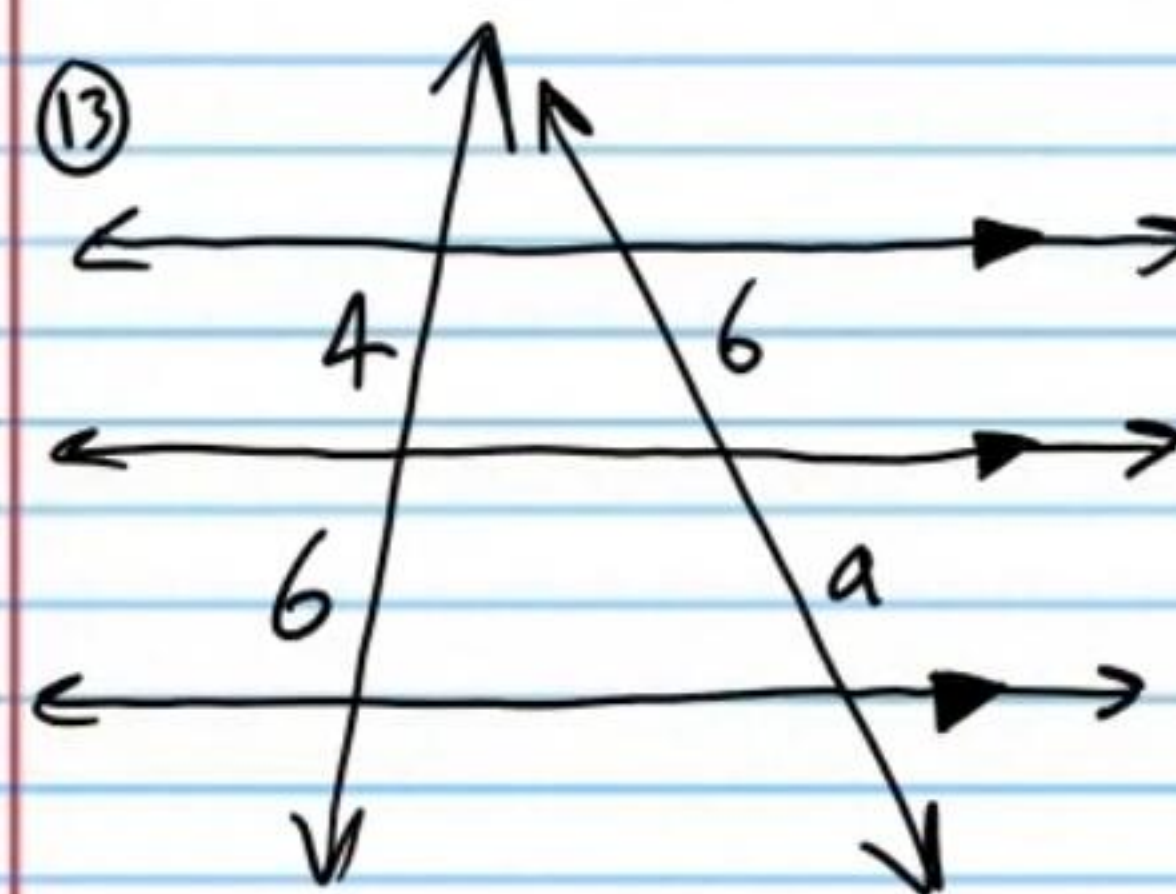


Transversal Proportionality Theorem: If three parallel lines intersect two transversals, then they divide the transversals proportionally.

If $\overleftrightarrow{AD} \parallel \overleftrightarrow{BE}$ and $\overleftrightarrow{BE} \parallel \overleftrightarrow{CF}$,
then $\frac{AB}{BC} = \frac{DE}{EF}$.



Find the unknown values.

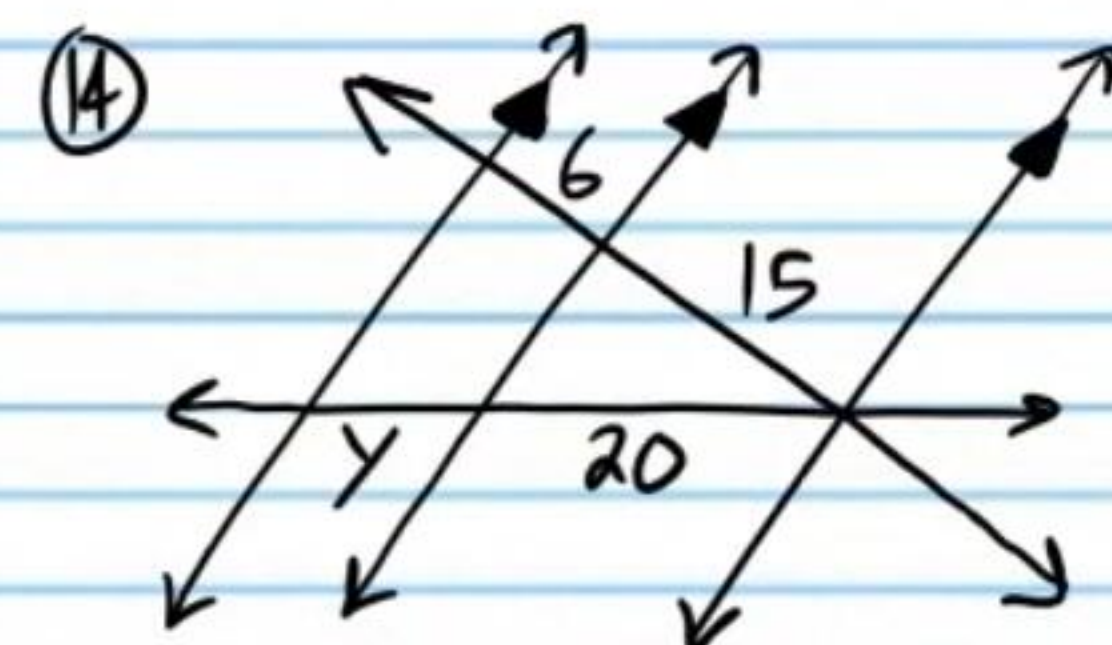


$$\frac{4}{6} = \frac{6}{a}$$

$$\frac{2}{3} = \frac{6}{a}$$

$$2a = 18$$

$$a = 9$$

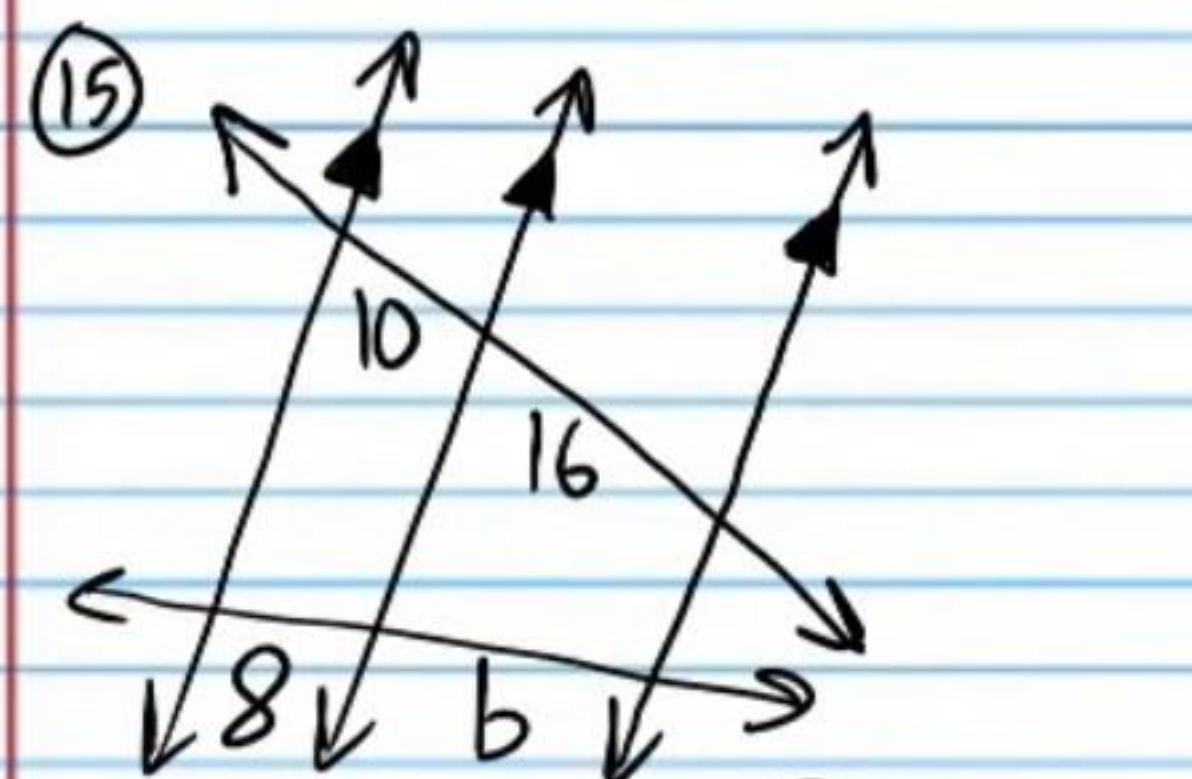


$$\frac{6}{15} = \frac{y}{20}$$

$$\frac{2}{5} = \frac{y}{20}$$

$$40 = 5y$$

$$8 = y$$

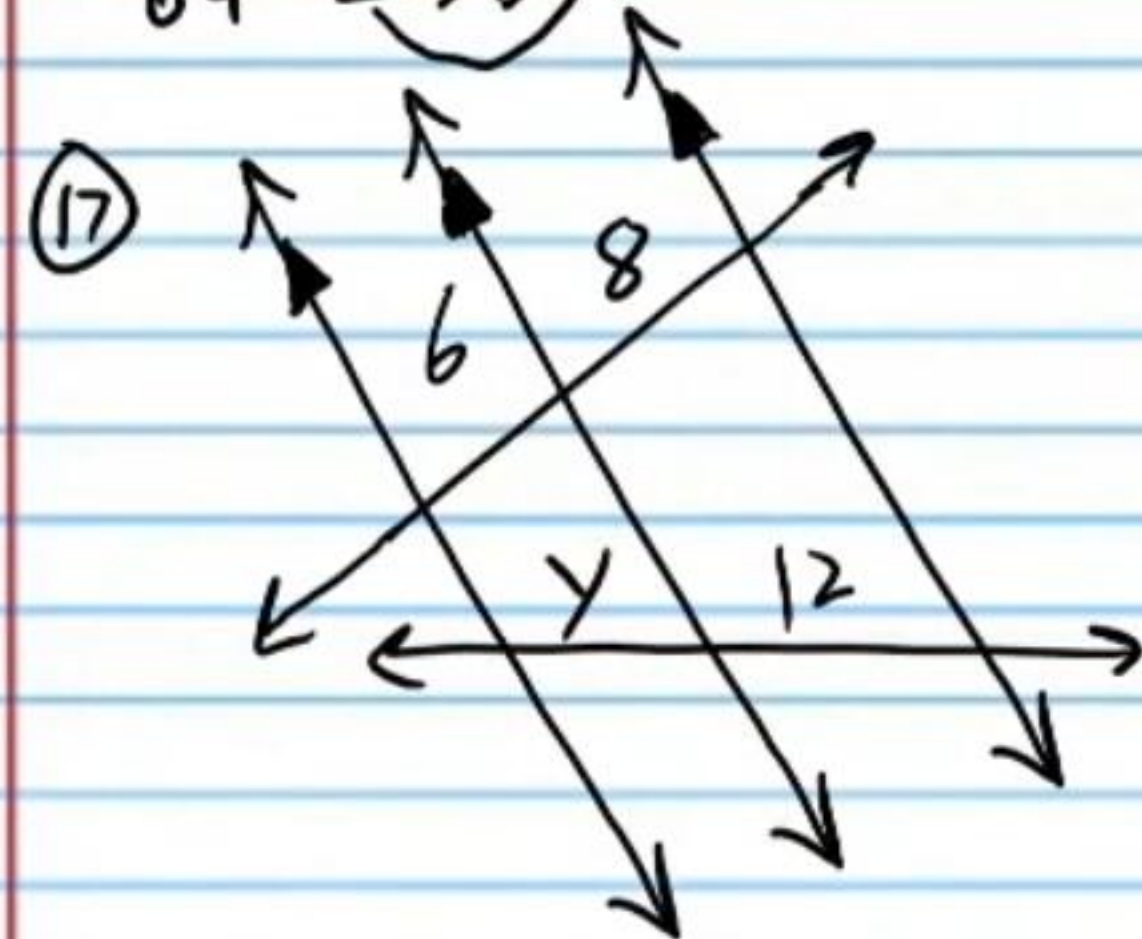


$$\frac{8}{b} = \frac{10}{16}$$

$$\frac{8}{b} = \frac{5}{8}$$

$$64 = 5b$$

$$\boxed{b = 12.8}$$

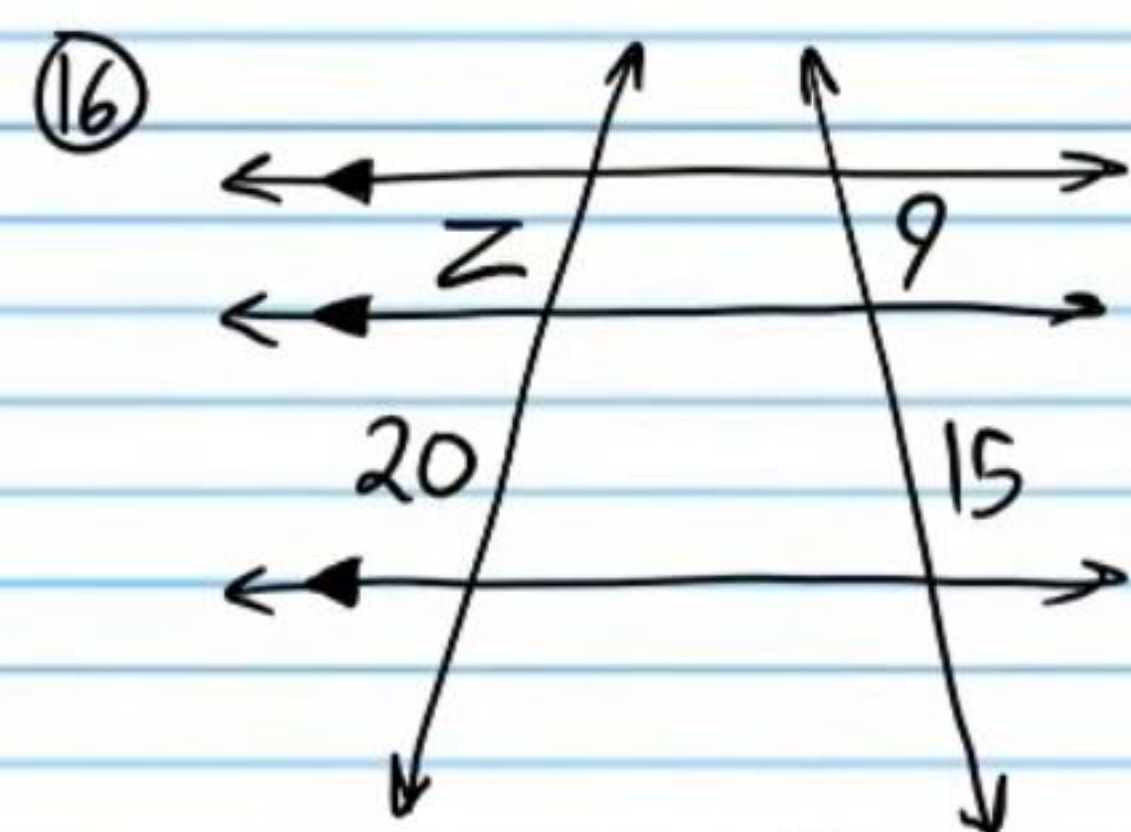


$$\frac{6}{8} = \frac{y}{12}$$

$$\frac{3}{4} = \frac{y}{12}$$

$$36 = 4y$$

$$\boxed{9 = y}$$

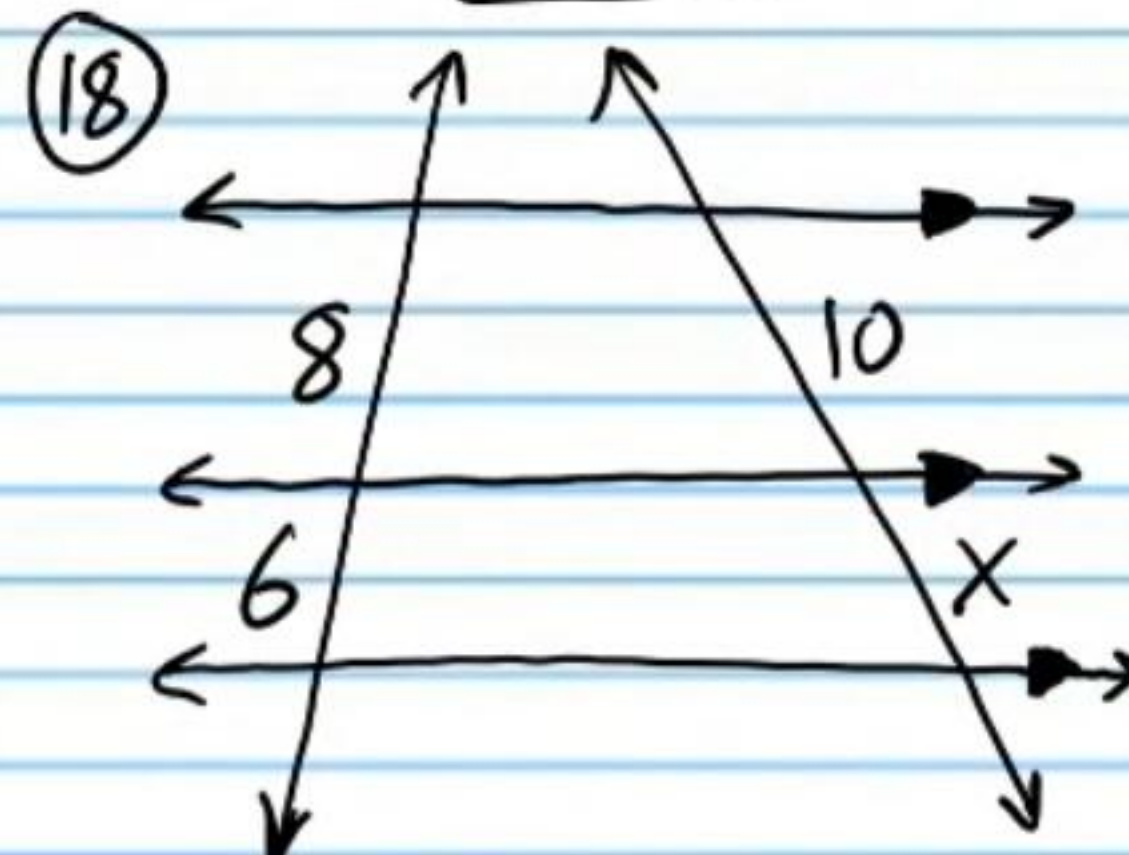


$$\frac{z}{20} = \frac{9}{15}$$

$$\frac{z}{20} = \frac{3}{5}$$

$$5z = 60$$

$$\boxed{z = 12}$$



$$\frac{8}{6} = \frac{10}{x}$$

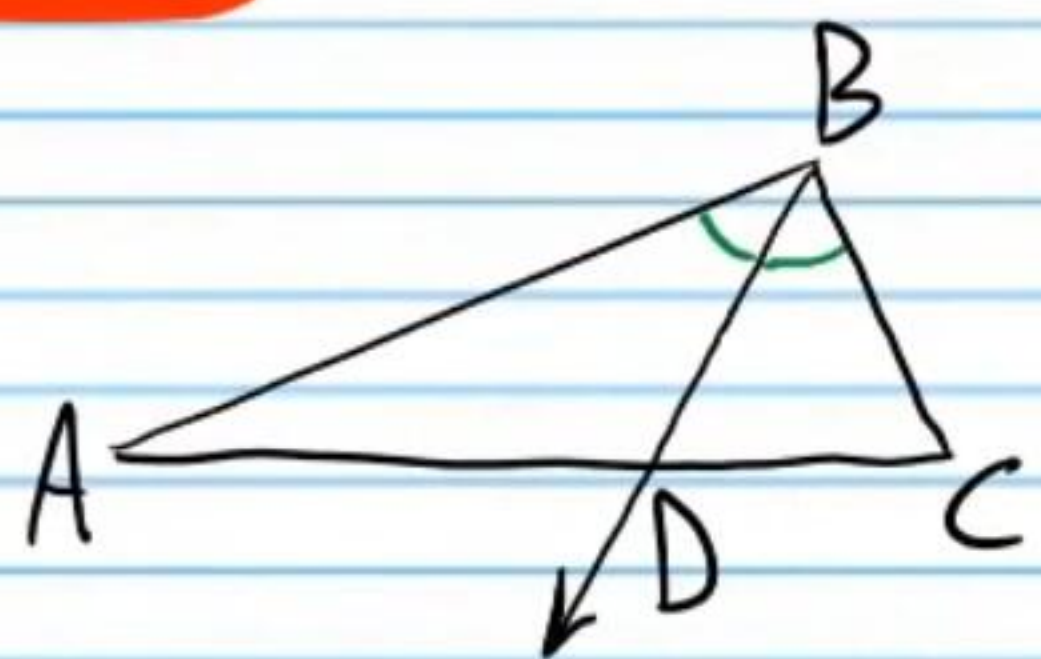
$$\frac{4}{3} = \frac{10}{x}$$

$$4x = 30$$

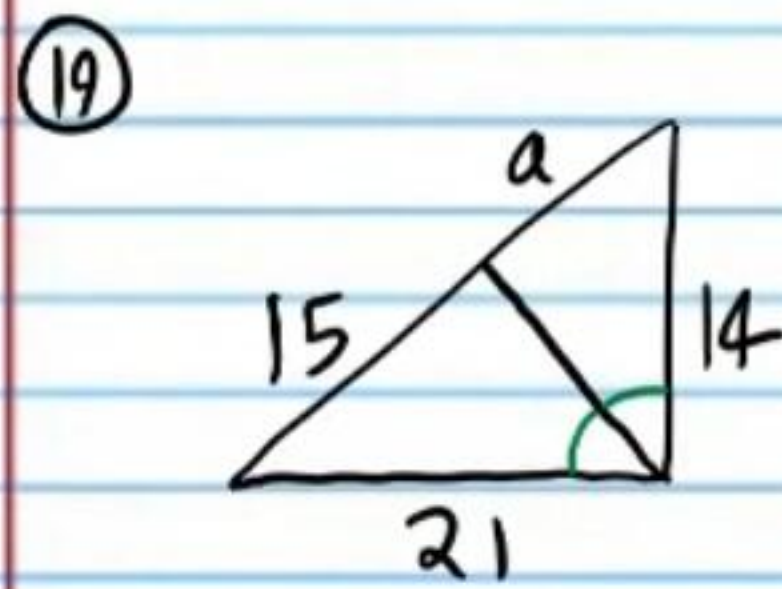
$$\boxed{x = 7.5}$$

Angle Bisector Theorem: If a ray bisects an angle of a triangle, then it divides the opposite side into segments whose lengths are proportional to the lengths of the other two sides.

If $m\angle ABD = m\angle CBD$,
then $\frac{AD}{DC} = \frac{AB}{BC}$.



Find the unknown values.

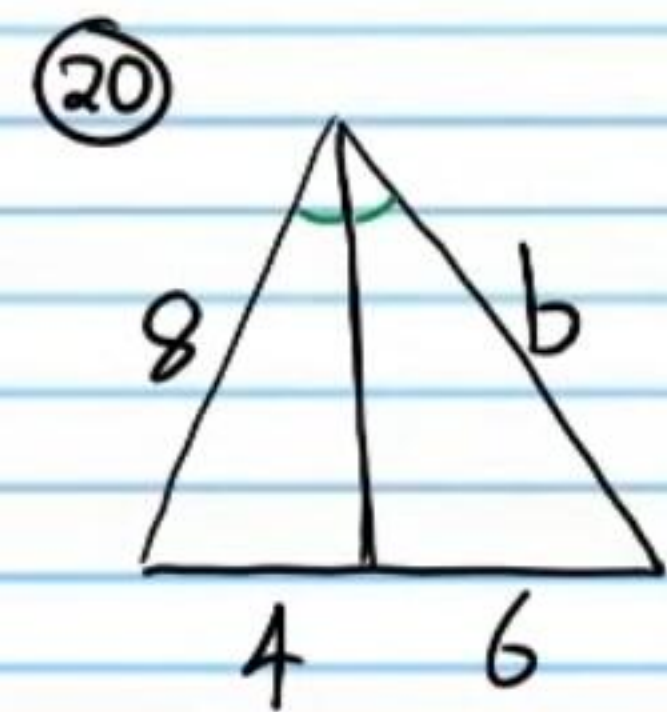


$$\frac{a}{15} = \frac{14}{21}$$

$$\frac{a}{15} = \frac{2}{3}$$

$$3a = 30$$

$$\boxed{a = 10}$$

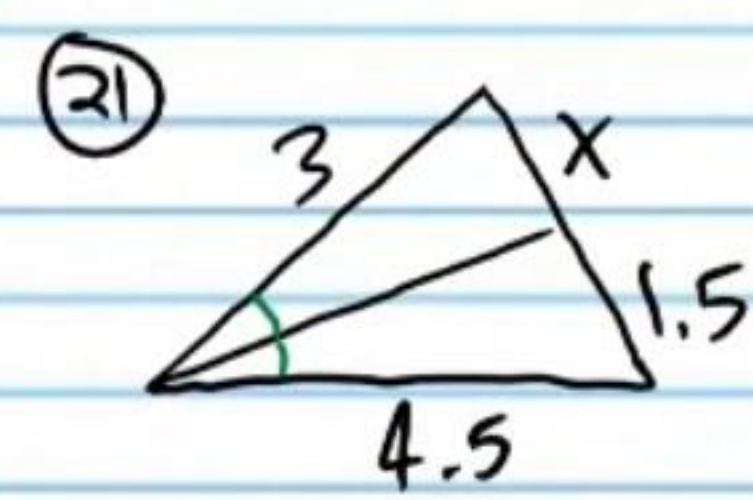


$$\frac{4}{6} = \frac{8}{b}$$

$$\frac{2}{3} = \frac{8}{b}$$

$$2b = 24$$

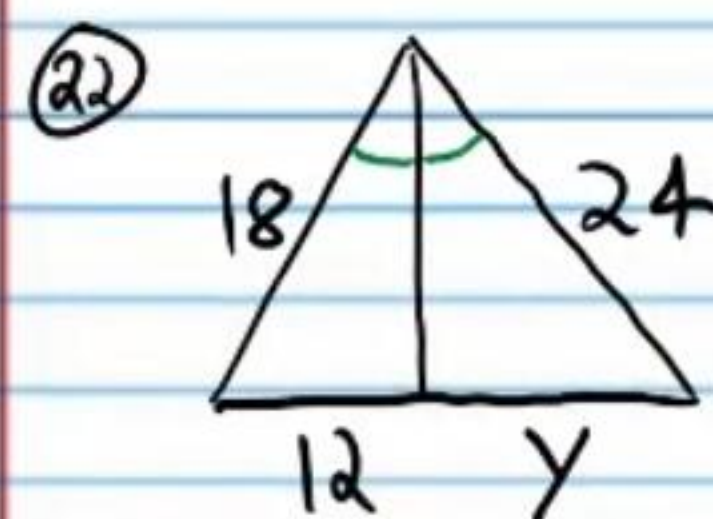
$$\boxed{b = 12}$$



$$\frac{x}{1.5} = \frac{3}{4.5}$$

$$4.5x = 4.5$$

$$\boxed{x = 1}$$

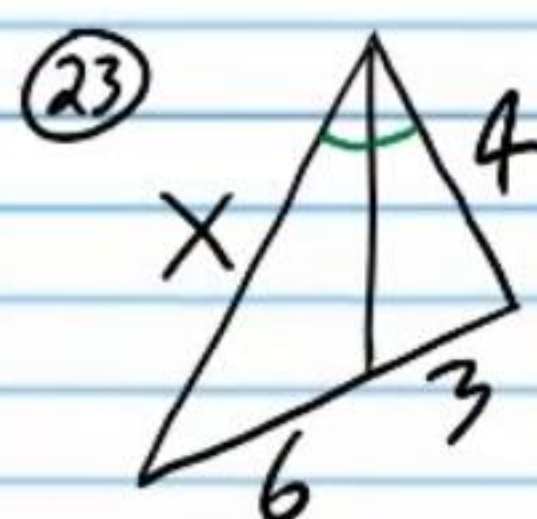


$$\frac{12}{y} = \frac{18}{24}$$

$$\frac{12}{y} = \frac{3}{4}$$

$$48 = 3y$$

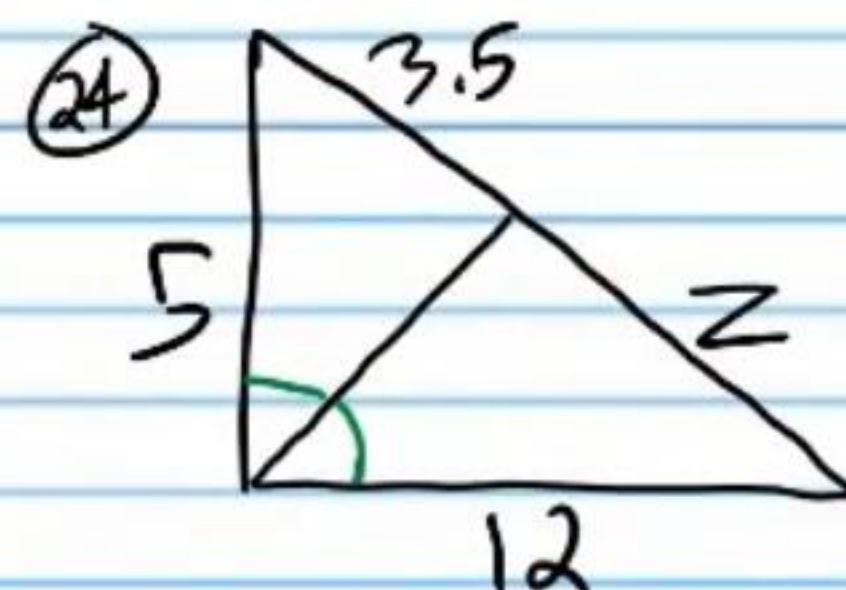
$$\boxed{16 = y}$$



$$\frac{6}{3} = \frac{x}{4}$$

$$\frac{2}{1} = \frac{x}{4}$$

$$\boxed{8 = x}$$



$$\frac{3.5}{z} = \frac{5}{12}$$

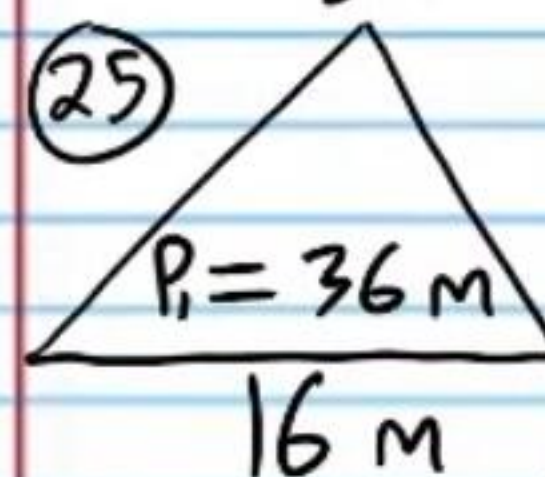
$$42 = 5z$$

$$\boxed{8.4 = z}$$

Perimeter and Area of Similar Polygons

Proportional Perimeters Theorem: If the ratio of corresponding sides of two similar polygons is a/b , then the ratio of the perimeters of the polygons is also a/b .

If the following polygons are similar, find the missing perimeters.



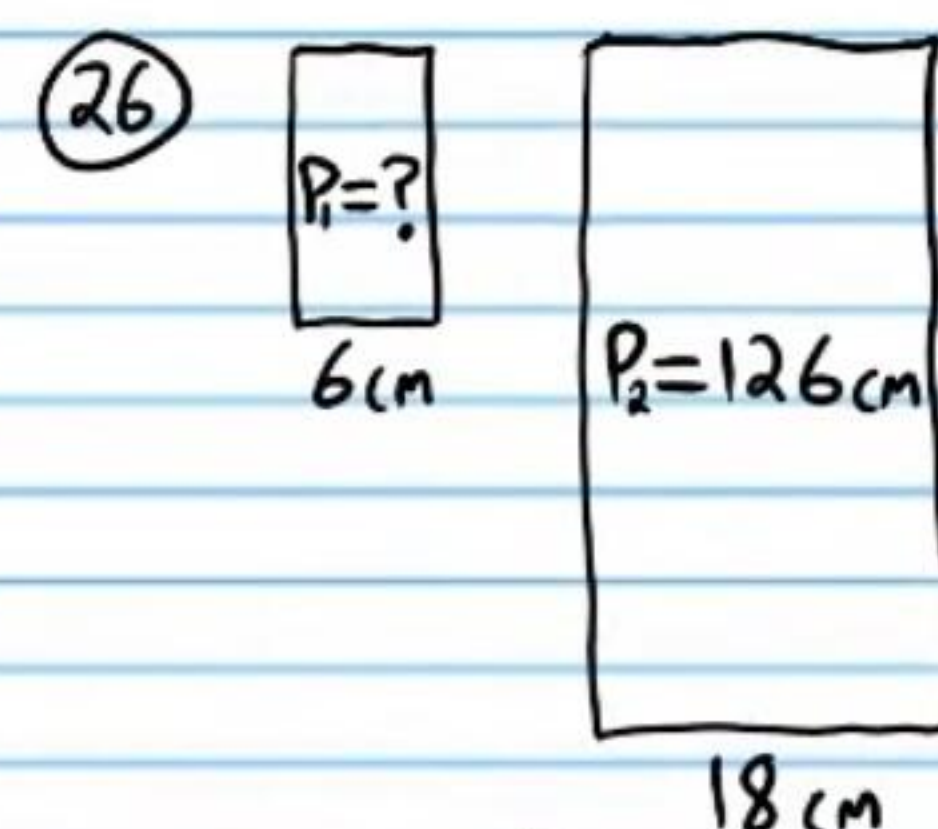
$$\frac{16}{20} = \frac{P_1}{P_2}$$

$$\frac{16}{20} = \frac{36}{P_2}$$

$$\frac{4}{5} = \frac{36}{P_2}$$

$$4P_2 = 180$$

$$\boxed{P_2 = 45 \text{ m}}$$



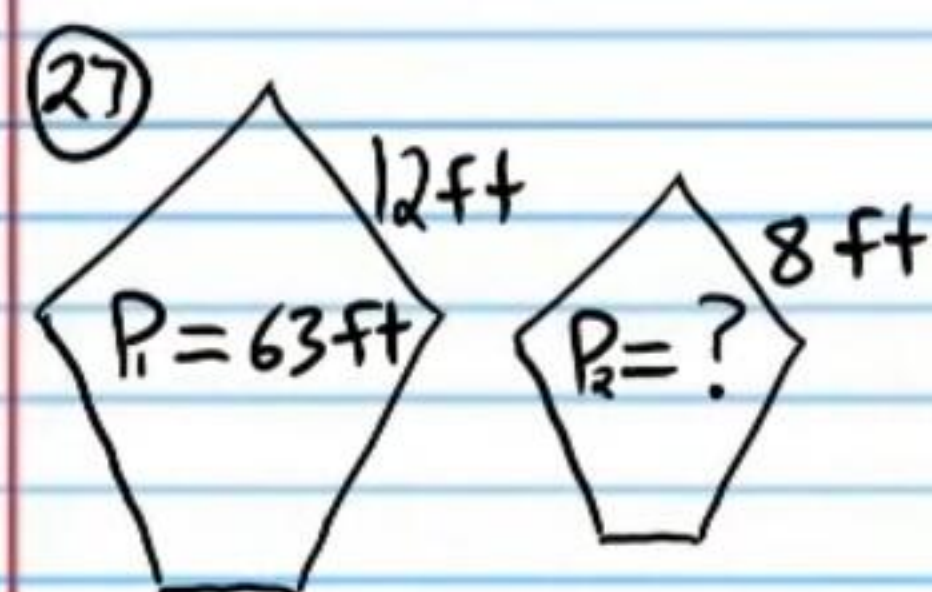
$$\frac{6}{18} = \frac{P_1}{P_2}$$

$$\frac{6}{18} = \frac{P_1}{126}$$

$$\frac{1}{3} = \frac{P_1}{126}$$

$$126 = 3P_1$$

$$\boxed{42 \text{ cm} = P_1}$$

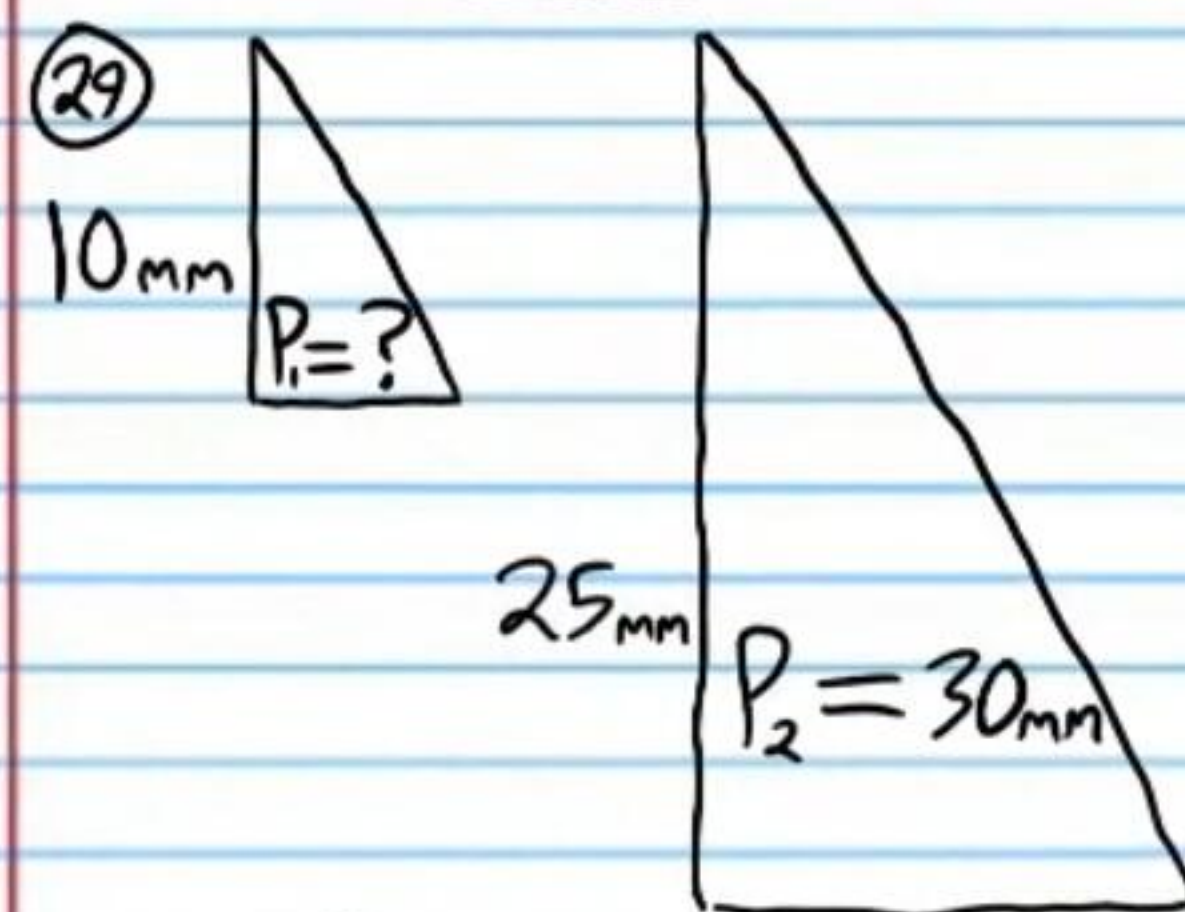


$$\frac{12}{8} = \frac{63}{P_2}$$

$$\frac{3}{2} = \frac{63}{P_2}$$

$$3P_2 = 126$$

$$P_2 = 42 \text{ ft}^2$$

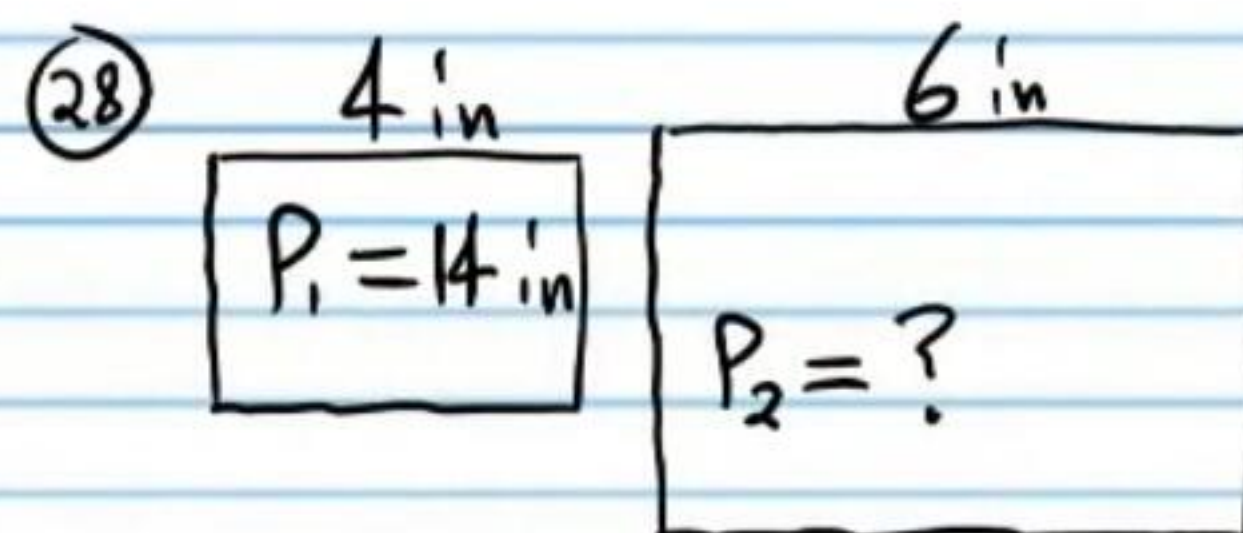


$$\frac{10}{25} = \frac{P_1}{30}$$

$$\frac{2}{5} = \frac{P_1}{30}$$

$$60 = 5P_1$$

$$12 \text{ mm}^2 = P_1$$

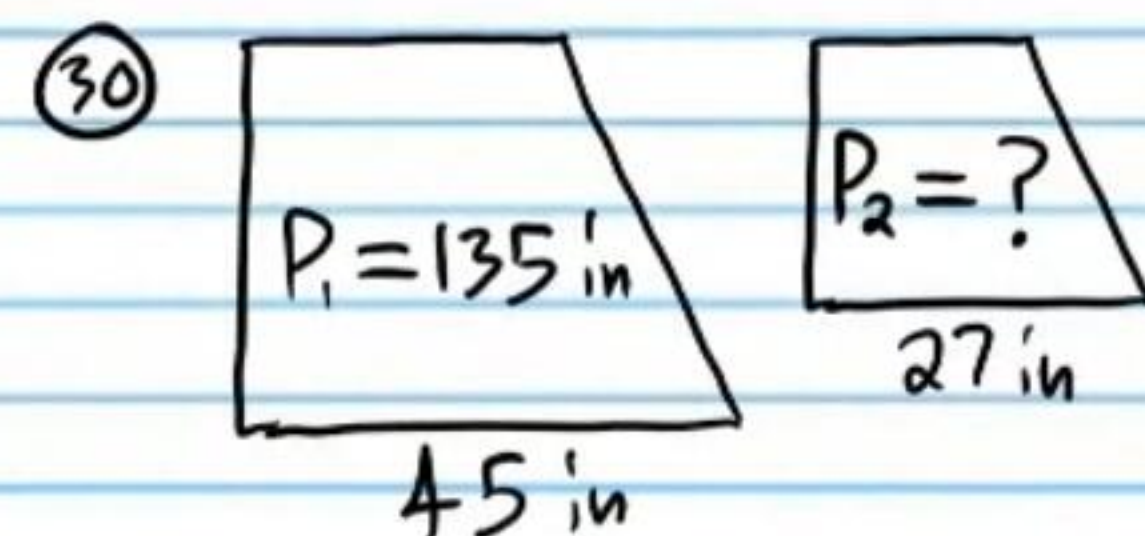


$$\frac{4}{6} = \frac{14}{P_2}$$

$$\frac{2}{3} = \frac{14}{P_2}$$

$$2P_2 = 42$$

$$P_2 = 21 \text{ in}^2$$



$$\frac{45}{27} = \frac{135}{P_2}$$

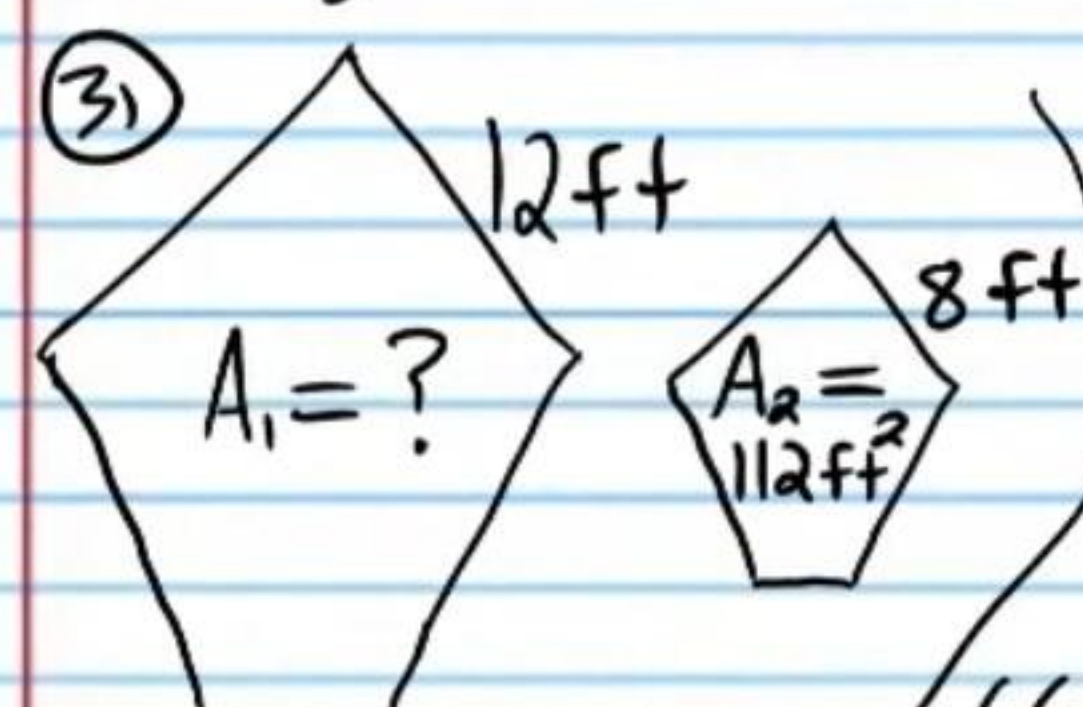
$$\frac{5}{3} = \frac{135}{P_2}$$

$$5P_2 = 405$$

$$P_2 = 81 \text{ in}^2$$

Proportional Areas Theorem: If the ratio of corresponding sides of two similar polygons is a/b , then the ratio of the areas of the polygons is $(a/b)^2$.

If the following polygons are similar, find the missing areas.



$$\left(\frac{12}{8}\right)^2 = \frac{A_1}{A_2}$$

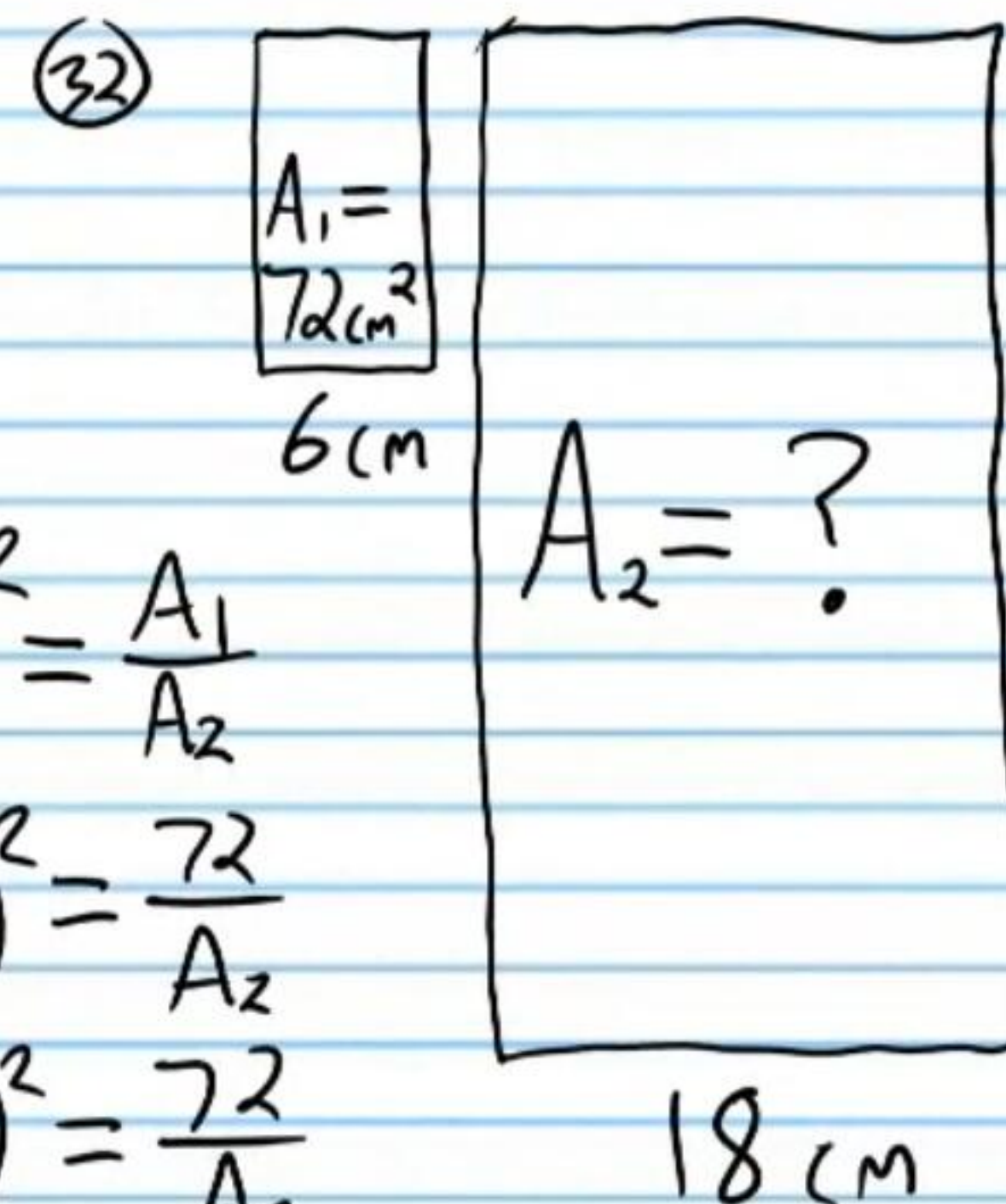
$$\left(\frac{12}{8}\right)^2 = \frac{A_1}{112}$$

$$\left(\frac{3}{2}\right)^2 = \frac{A_1}{112}$$

$$\frac{9}{4} = \frac{A_1}{112}$$

$$1,008 = 4A_1$$

$$252 \text{ ft}^2 = A_1$$



$$\left(\frac{6}{18}\right)^2 = \frac{A_1}{A_2}$$

$$\left(\frac{6}{18}\right)^2 = \frac{72}{A_2}$$

$$\left(\frac{1}{3}\right)^2 = \frac{72}{A_2}$$

$$\frac{1}{9} = \frac{72}{A_2}$$

$$A_2 = 648 \text{ cm}^2$$

33

$$\left(\frac{16}{20}\right)^2 = \frac{A_1}{160}$$

$$\left(\frac{4}{5}\right)^2 = \frac{A_1}{160}$$

$$\frac{16}{25} = \frac{A_1}{160}$$

$$2560 = 25A_1$$

$$102.4 \text{ m}^2 = A_1$$

34

$$\left(\frac{4}{6}\right)^2 = \frac{12}{A_2}$$

$$\left(\frac{2}{3}\right)^2 = \frac{12}{A_2}$$

$$\frac{4}{9} = \frac{12}{A_2}$$

$$4A_2 = 108$$

$$A_2 = 27 \text{ in}^2$$

35

$$\left(\frac{10}{25}\right)^2 = \frac{A_1}{175}$$

$$\left(\frac{2}{5}\right)^2 = \frac{A_1}{175}$$

$$\frac{4}{25} = \frac{A_1}{175}$$

$$700 = 25A_1$$

$$A_1 = 28 \text{ mm}^2$$

36

$$\left(\frac{45}{27}\right)^2 = \frac{A_1}{612}$$

$$\left(\frac{5}{3}\right)^2 = \frac{A_1}{612}$$

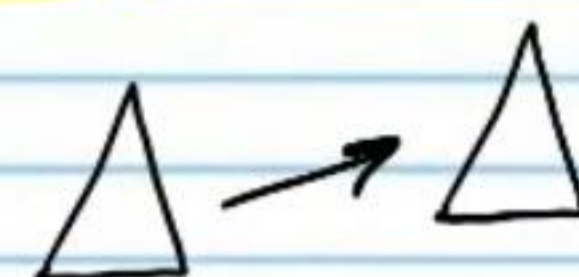
$$\frac{25}{9} = \frac{A_1}{612}$$

$$15,300 = 9A_1$$

$$A_1 = 1700 \text{ in}^2$$

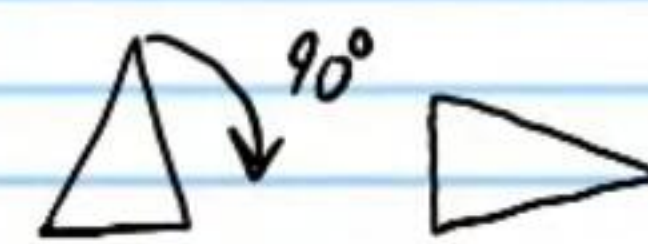
Transformations

I) Translation.



II) Rotation.

* There must be a center of rotation.



No Change in Size.

III) Reflection.

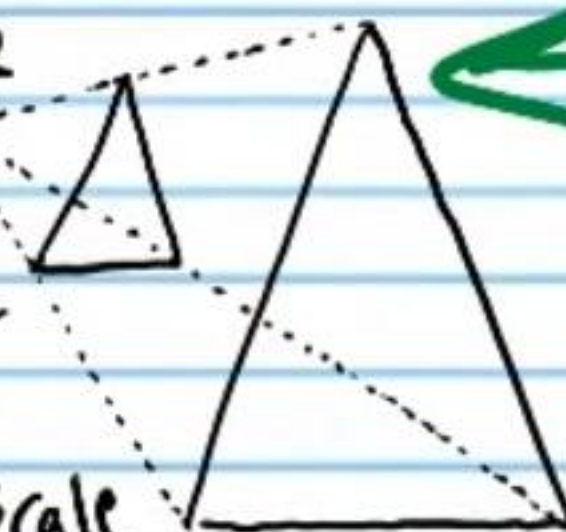
* There must be a line of reflection.



Congruence

IV) Dilation.

* There must be a center of dilation and a scale factor.

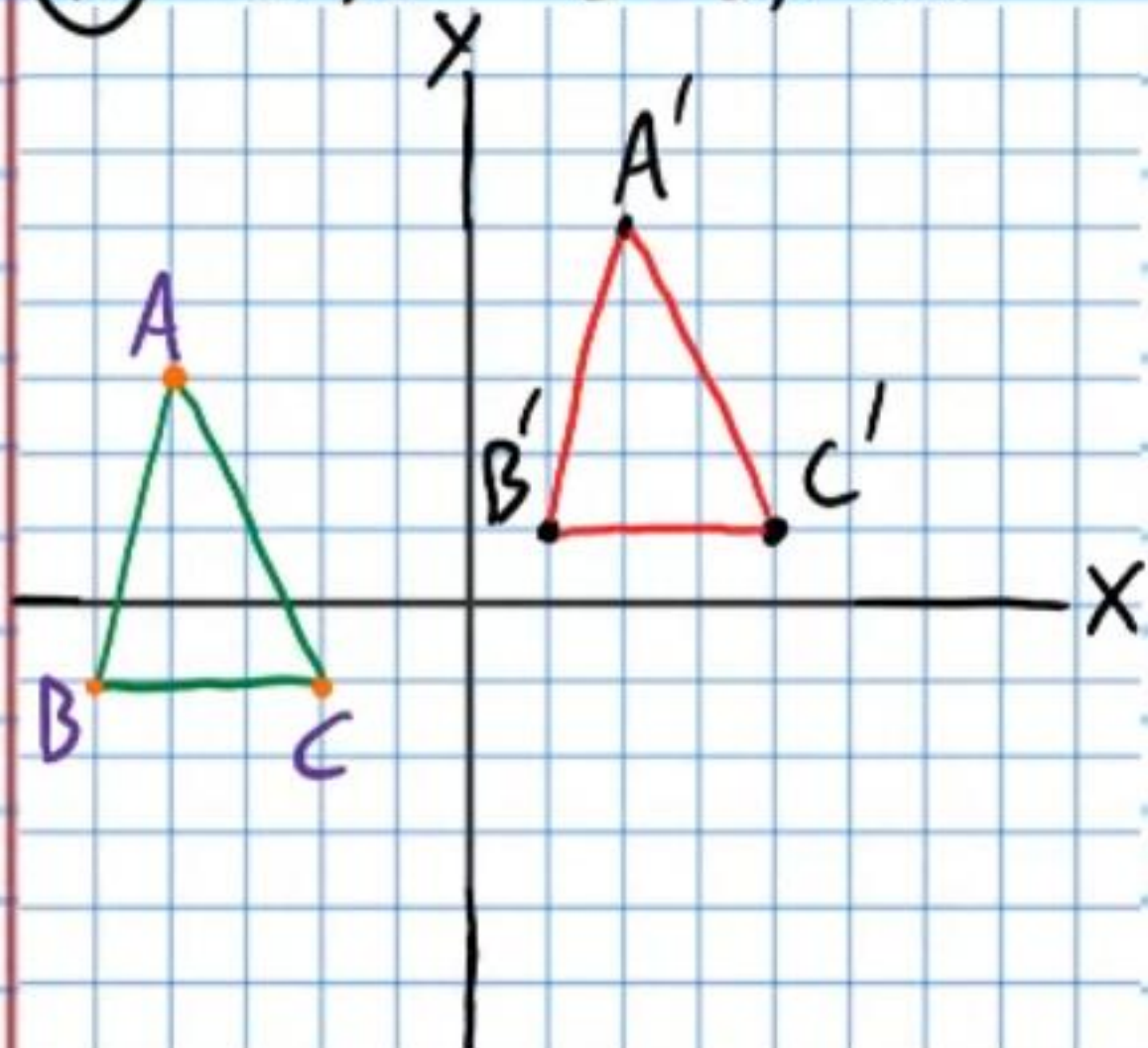


Size Changes.
Similarity

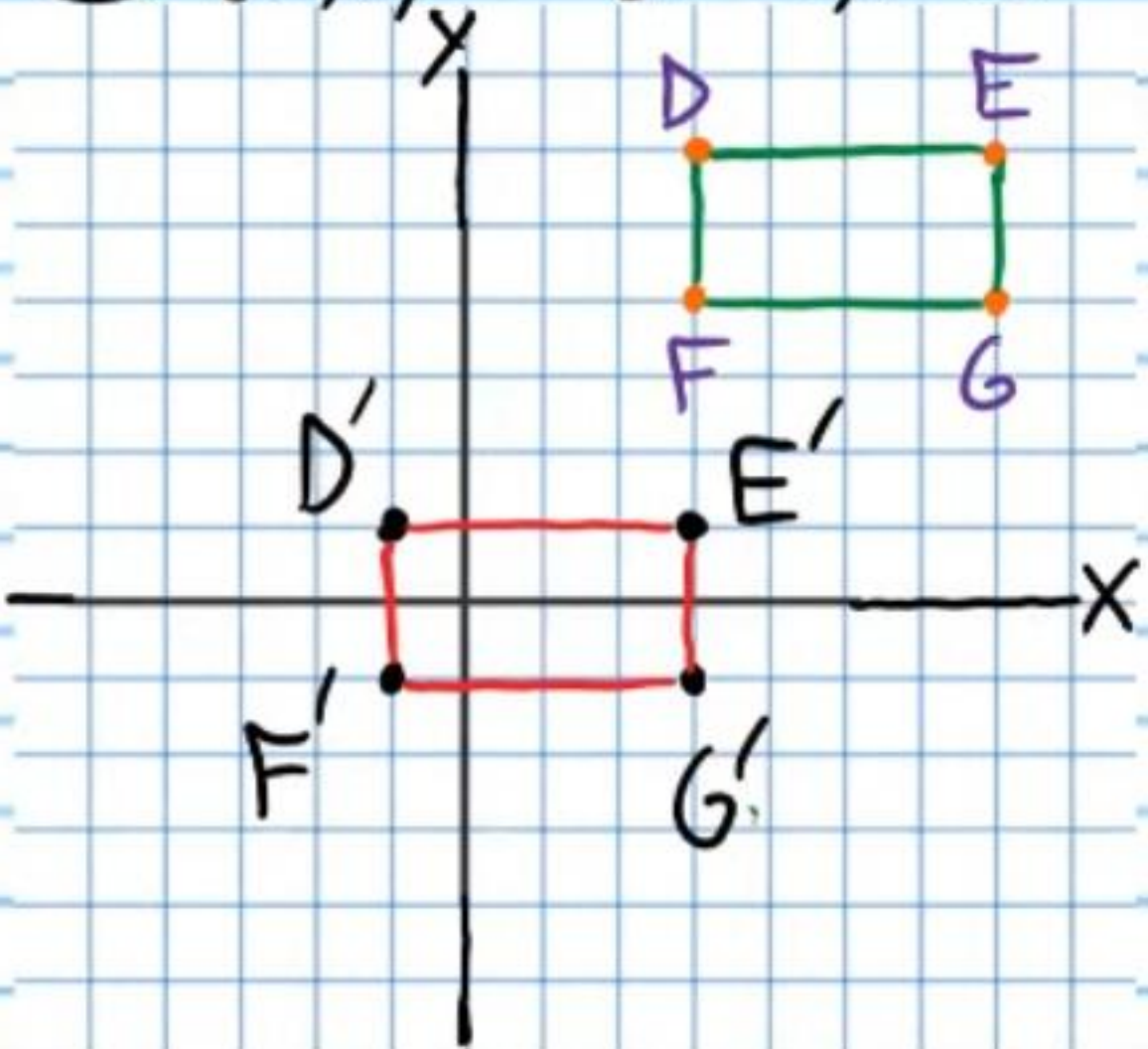
Translations

Given the "pre-image," graph the "image" using the given translation.

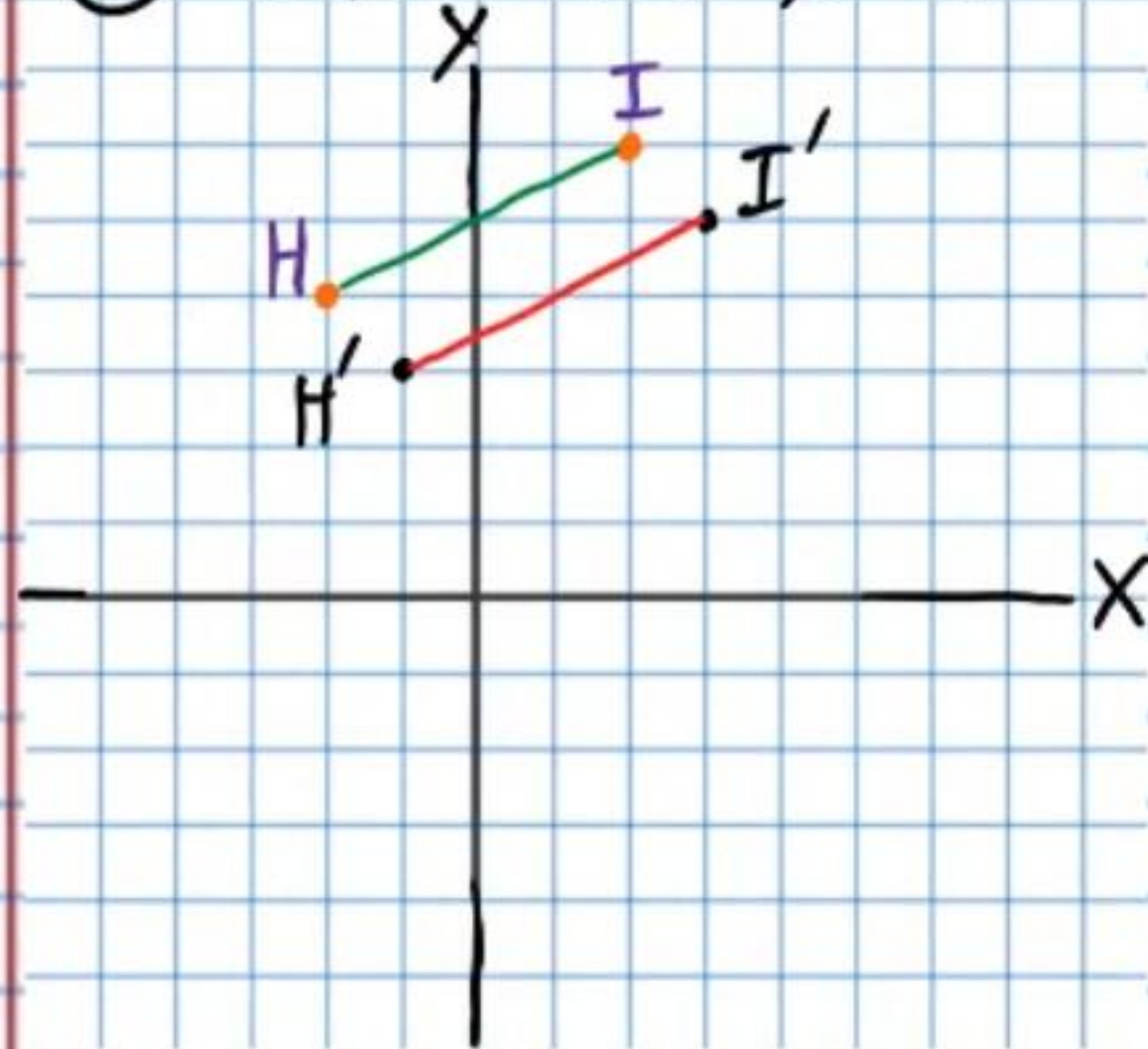
③⑦ $(x, y) \rightarrow (x+6, y+2)$



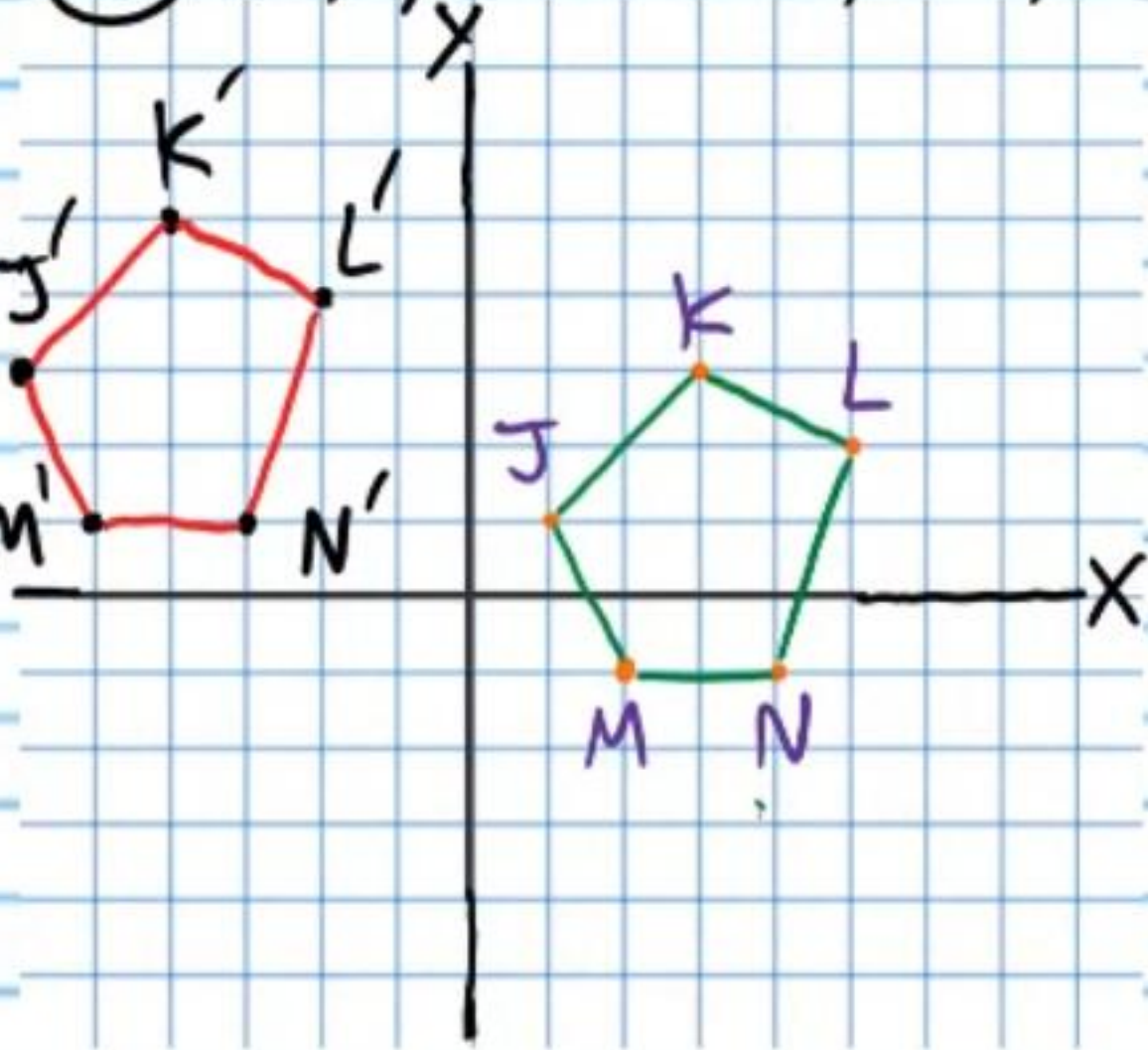
③⑧ $(x, y) \rightarrow (x-4, y-5)$



③⑨ $(x, y) \rightarrow (x+1, y-1)$



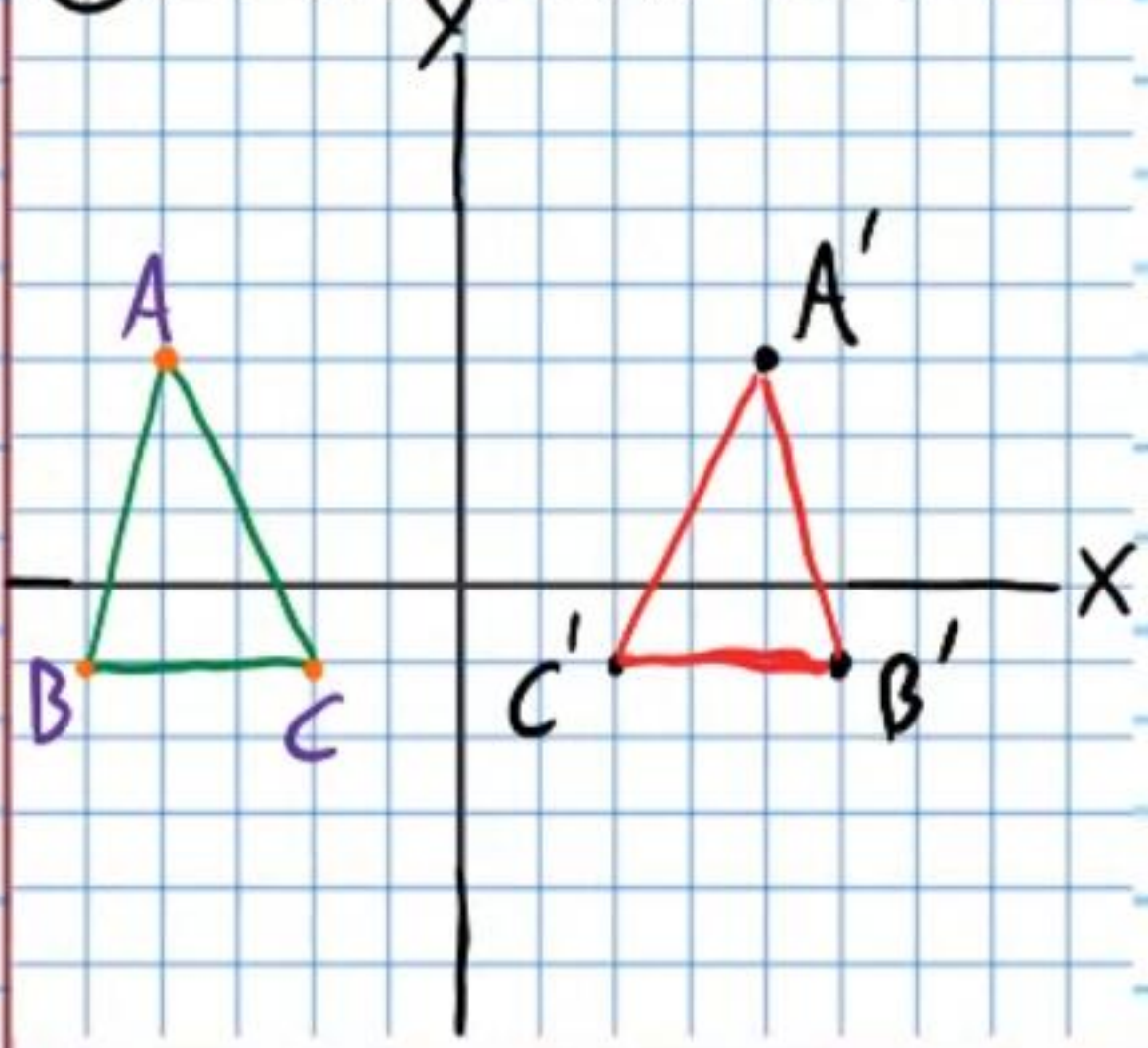
④⑩ $(x, y) \rightarrow (x-7, y+2)$



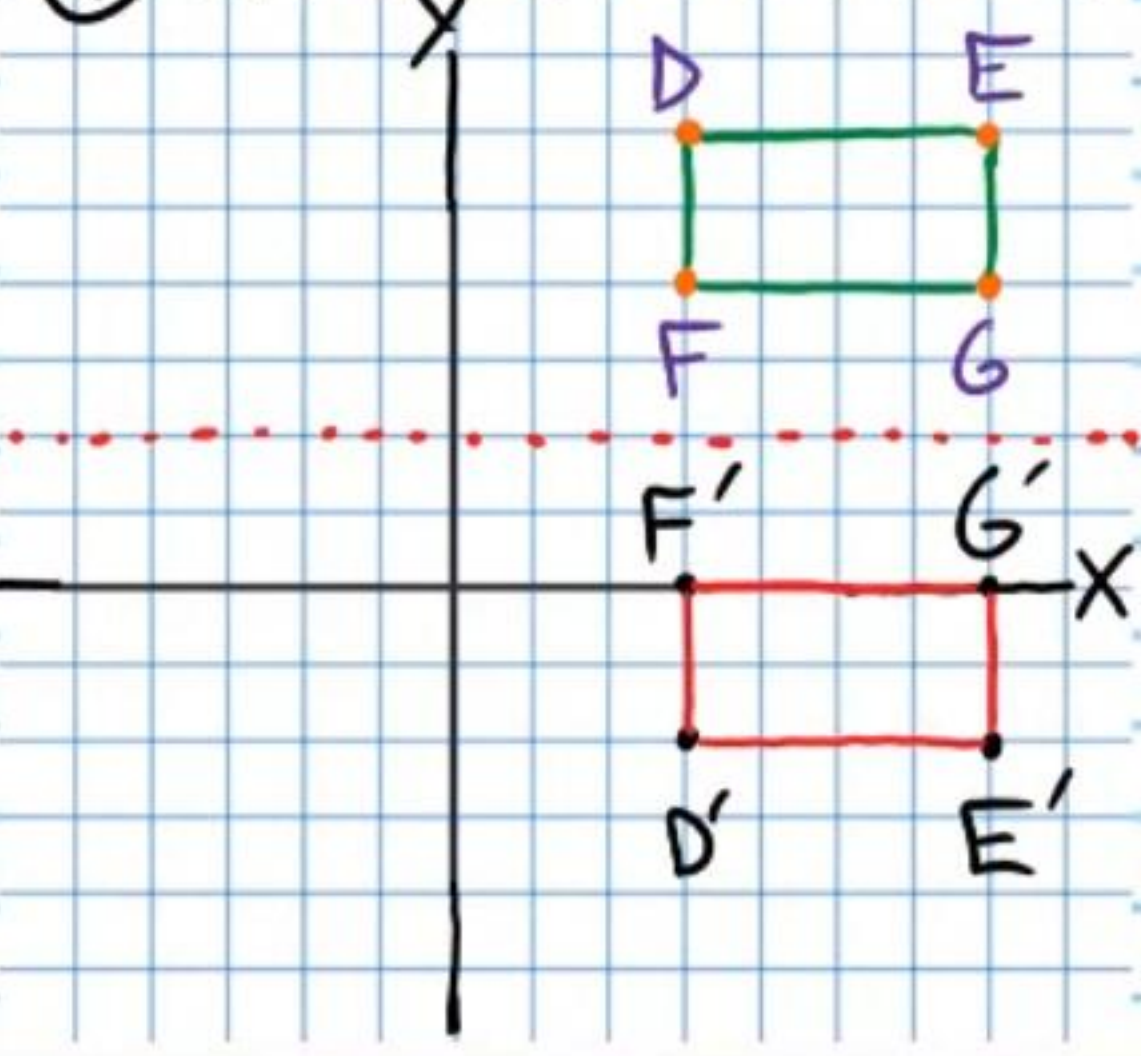
Reflections

Given the "pre-image," graph the "image" using the given reflection.

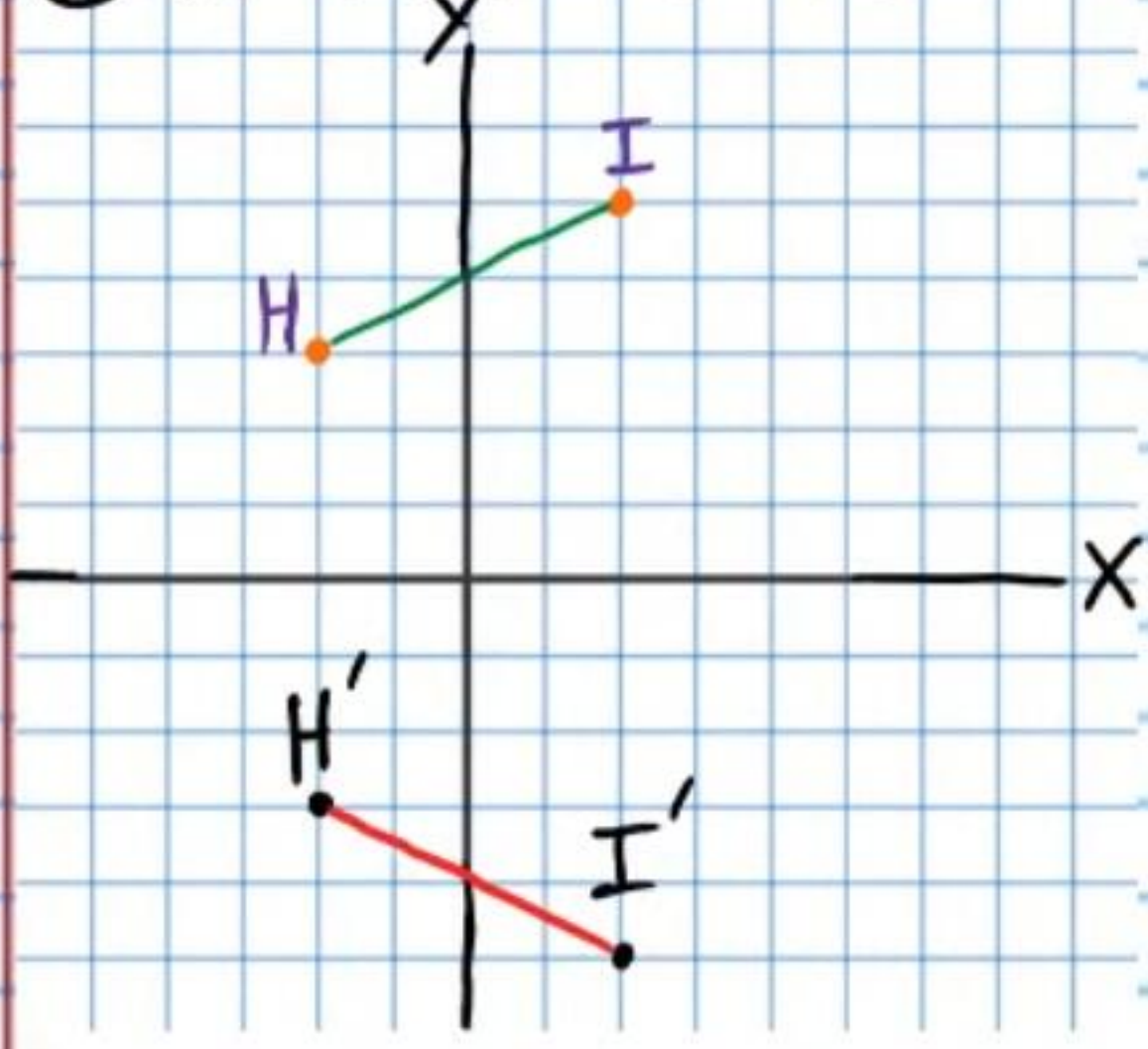
④① About the y -axis.



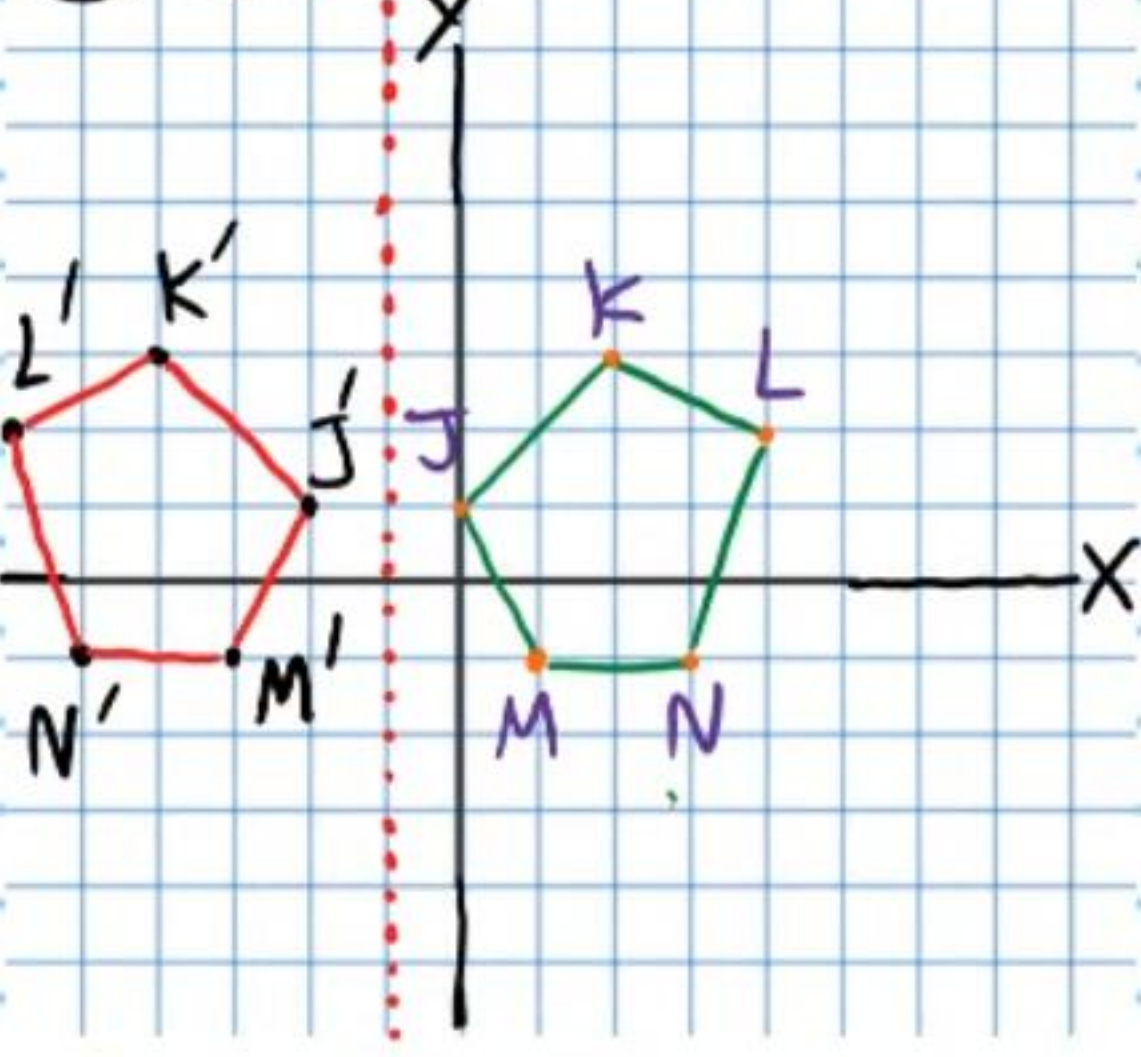
④② About the dotted line.



④③ About the x -axis.



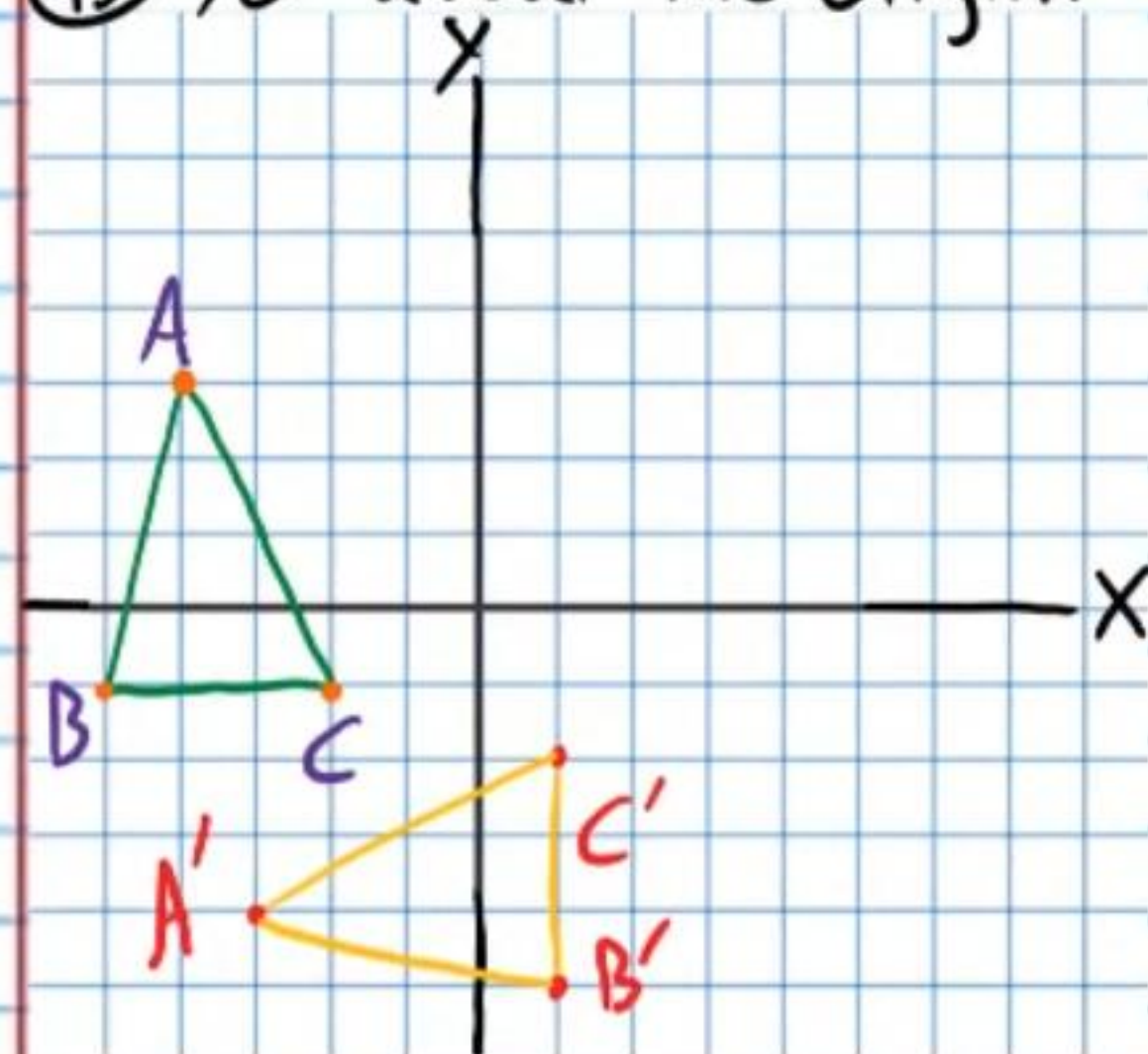
④④ About the dotted line.



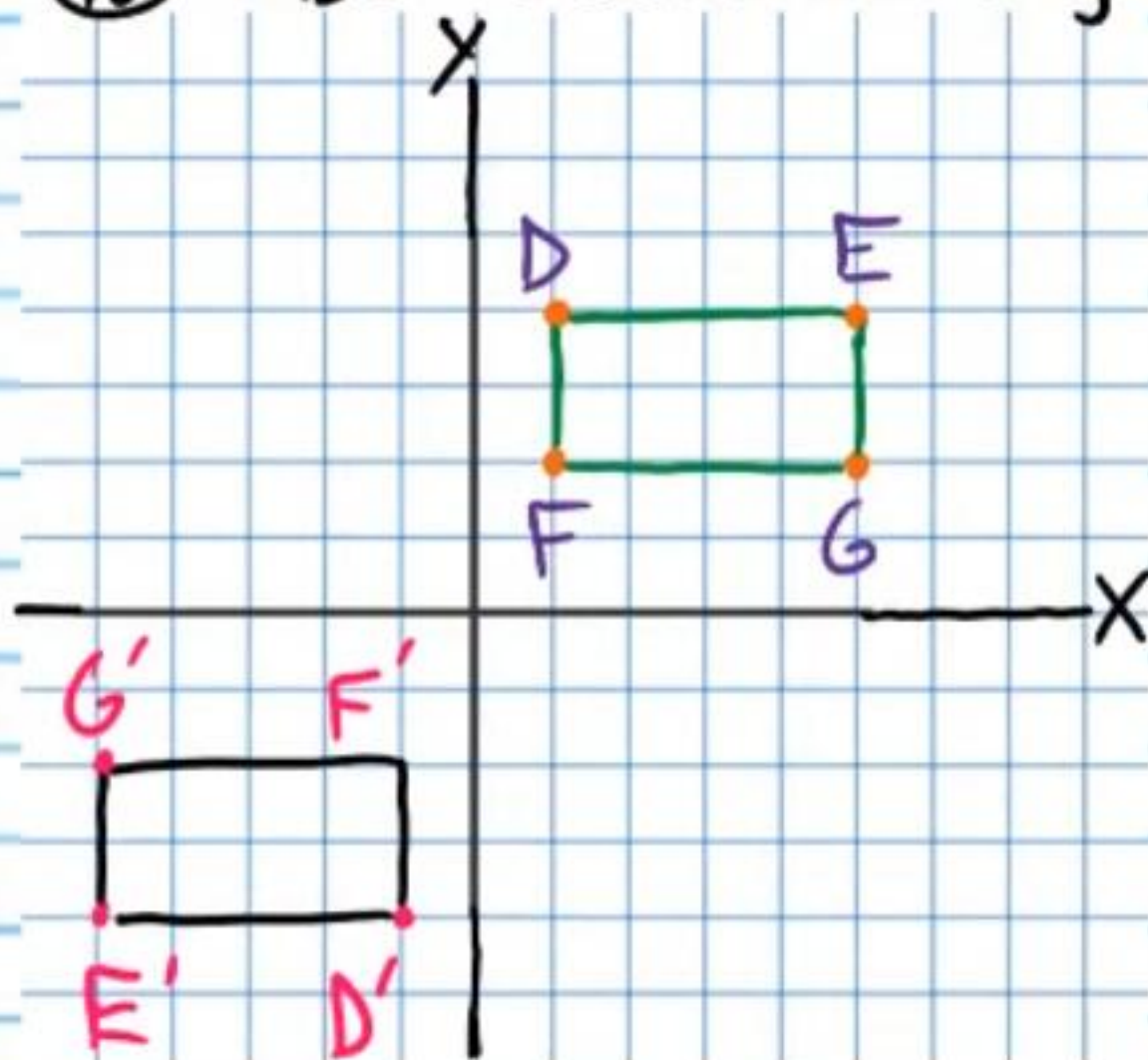
Rotations

Given the "pre-image," graph the "image" using the given rotation.

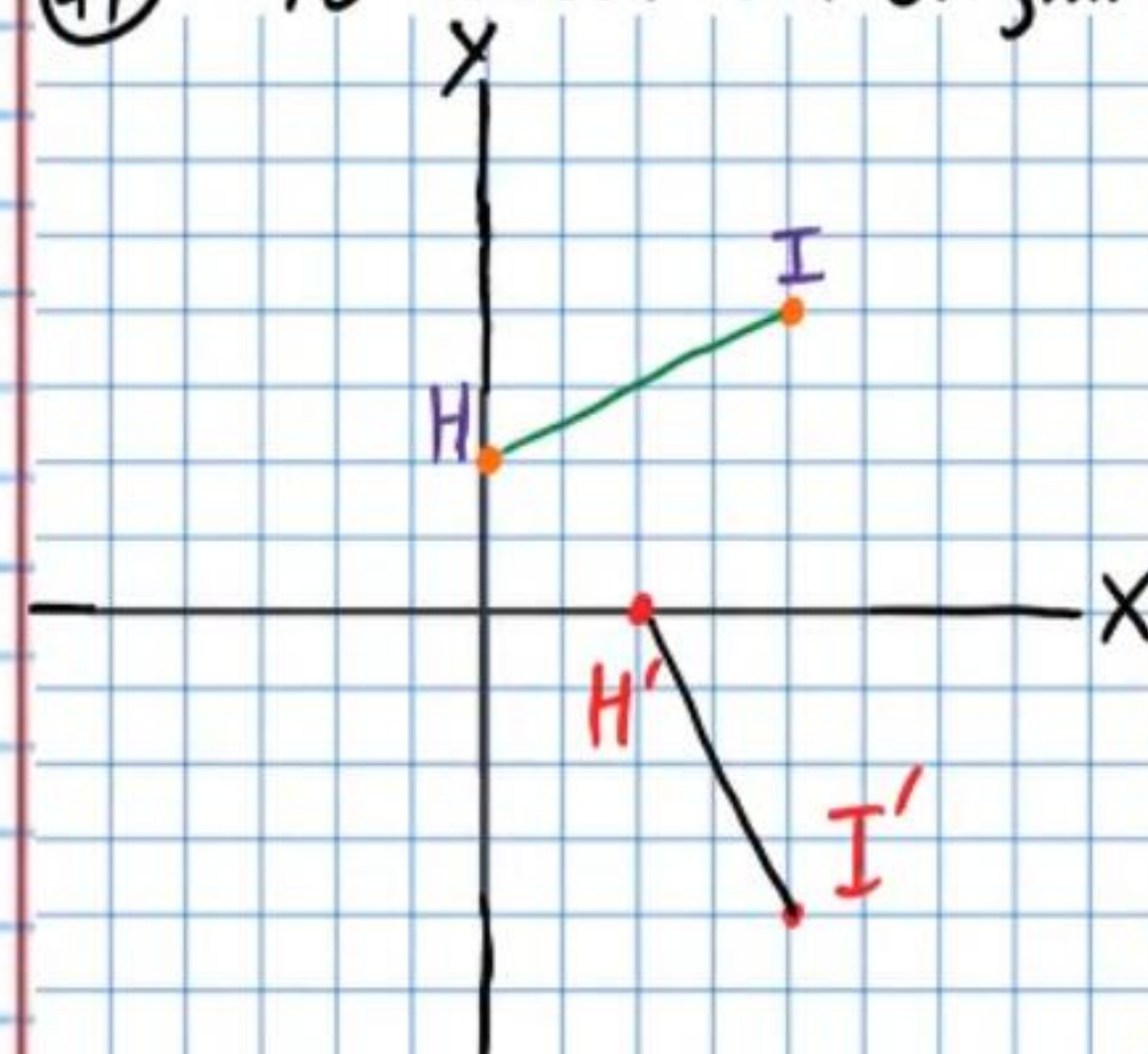
④⑤ 90° about the origin.



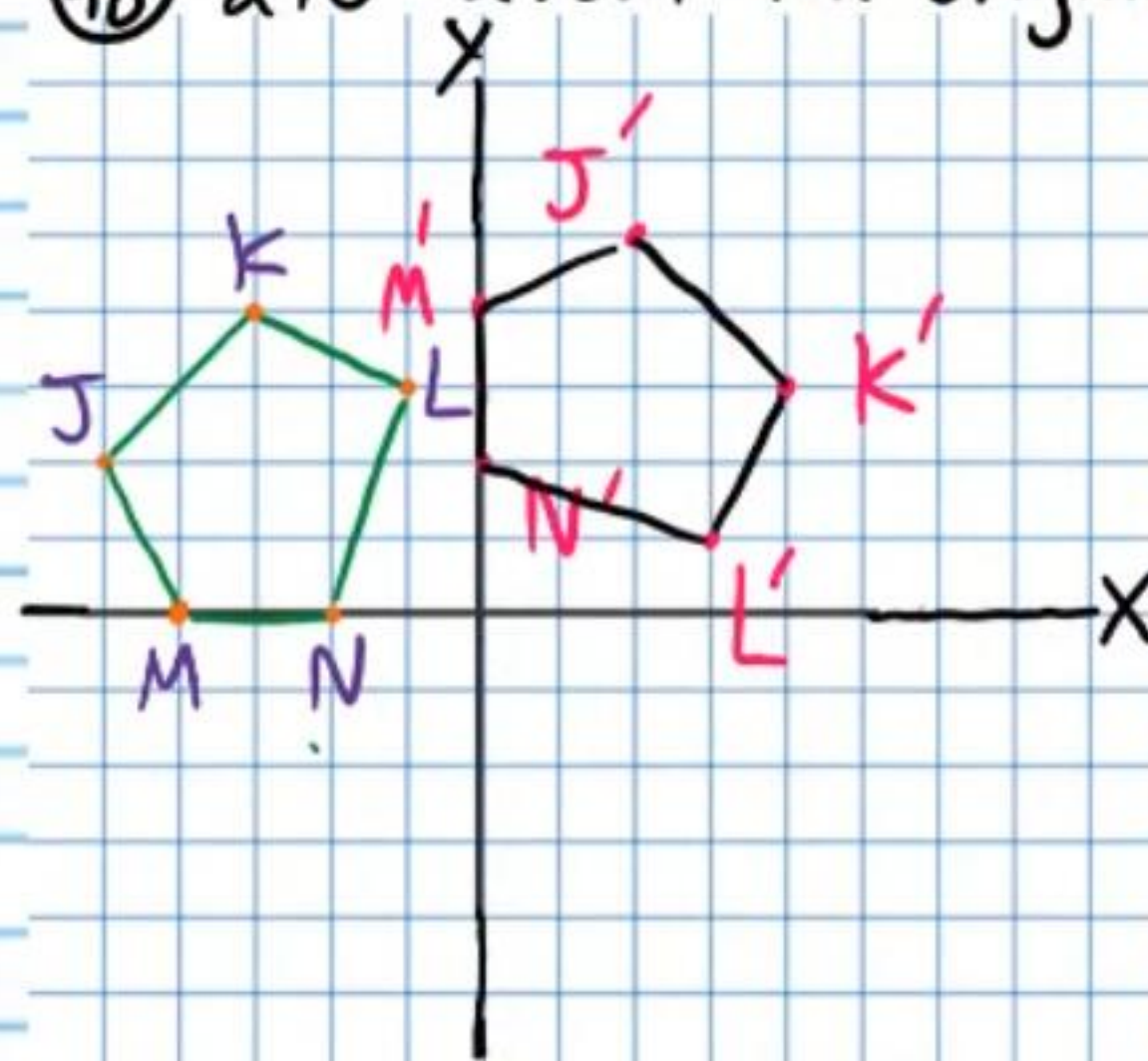
④⑥ -180° about the origin.



④⑦ -90° about the origin.



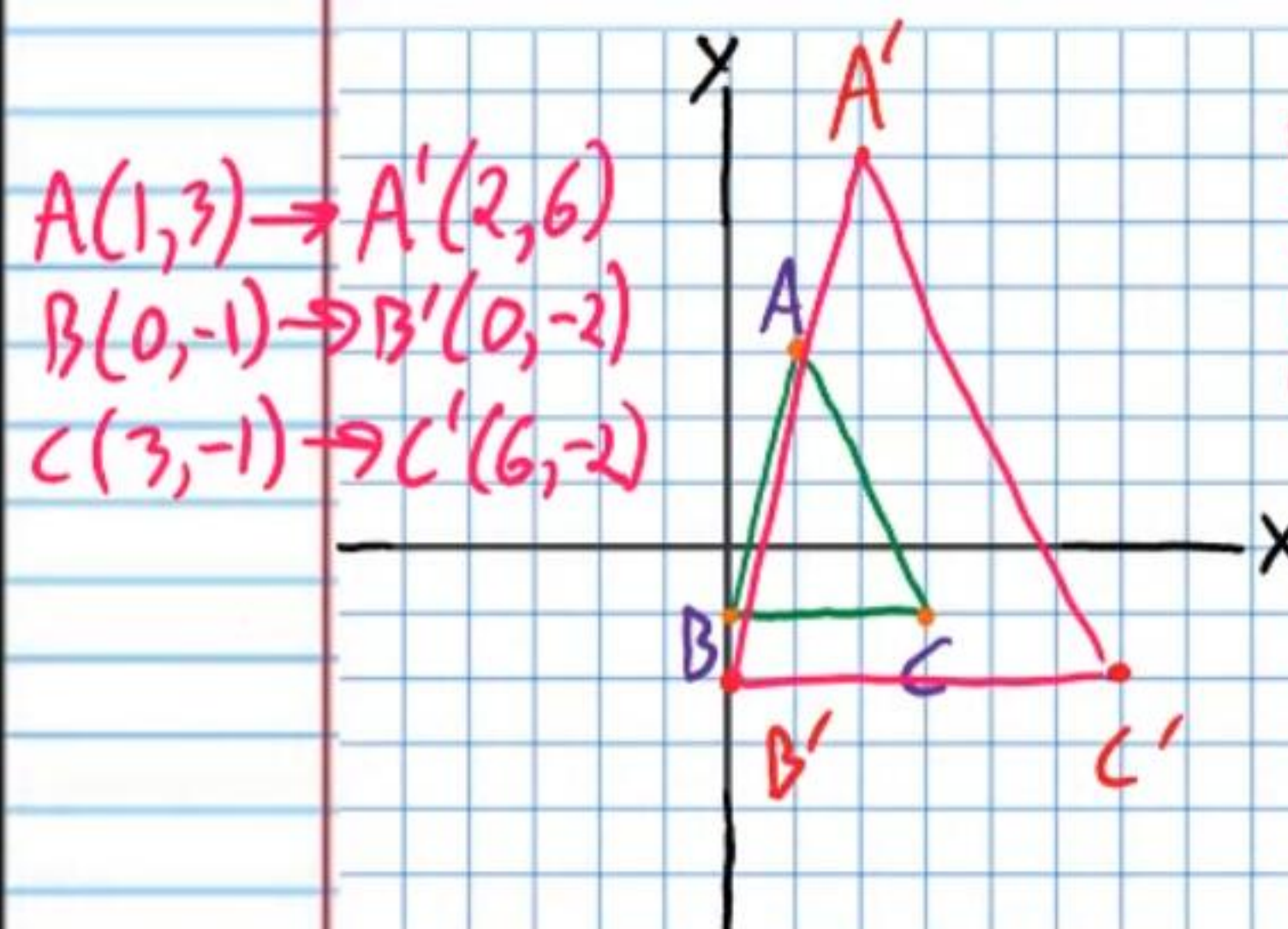
④⑧ 270° about the origin.



Dilations

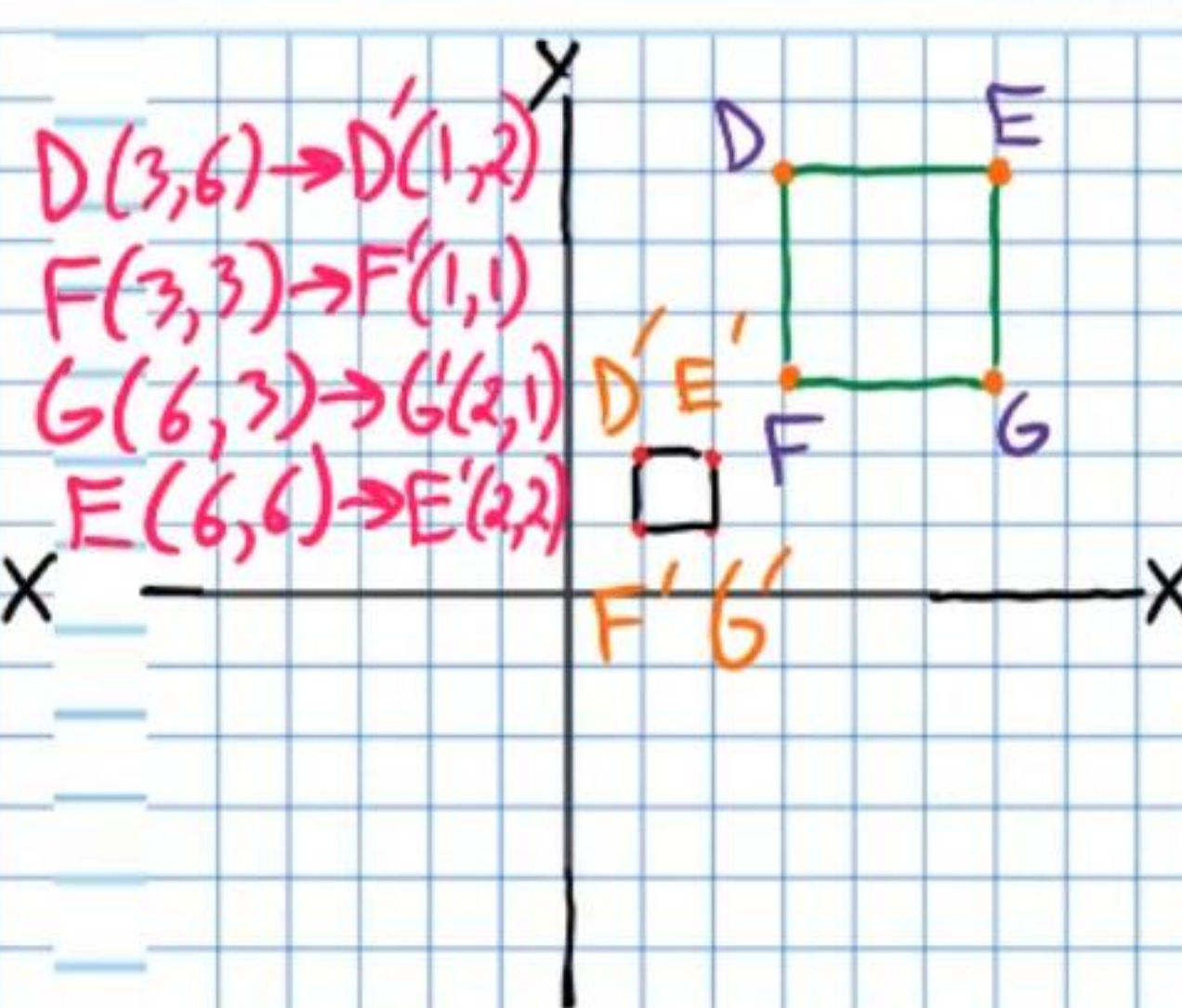
Given the "pre-image," graph the "image" using the given Dilation.

④⑨ About the origin with a scale factor of 2.



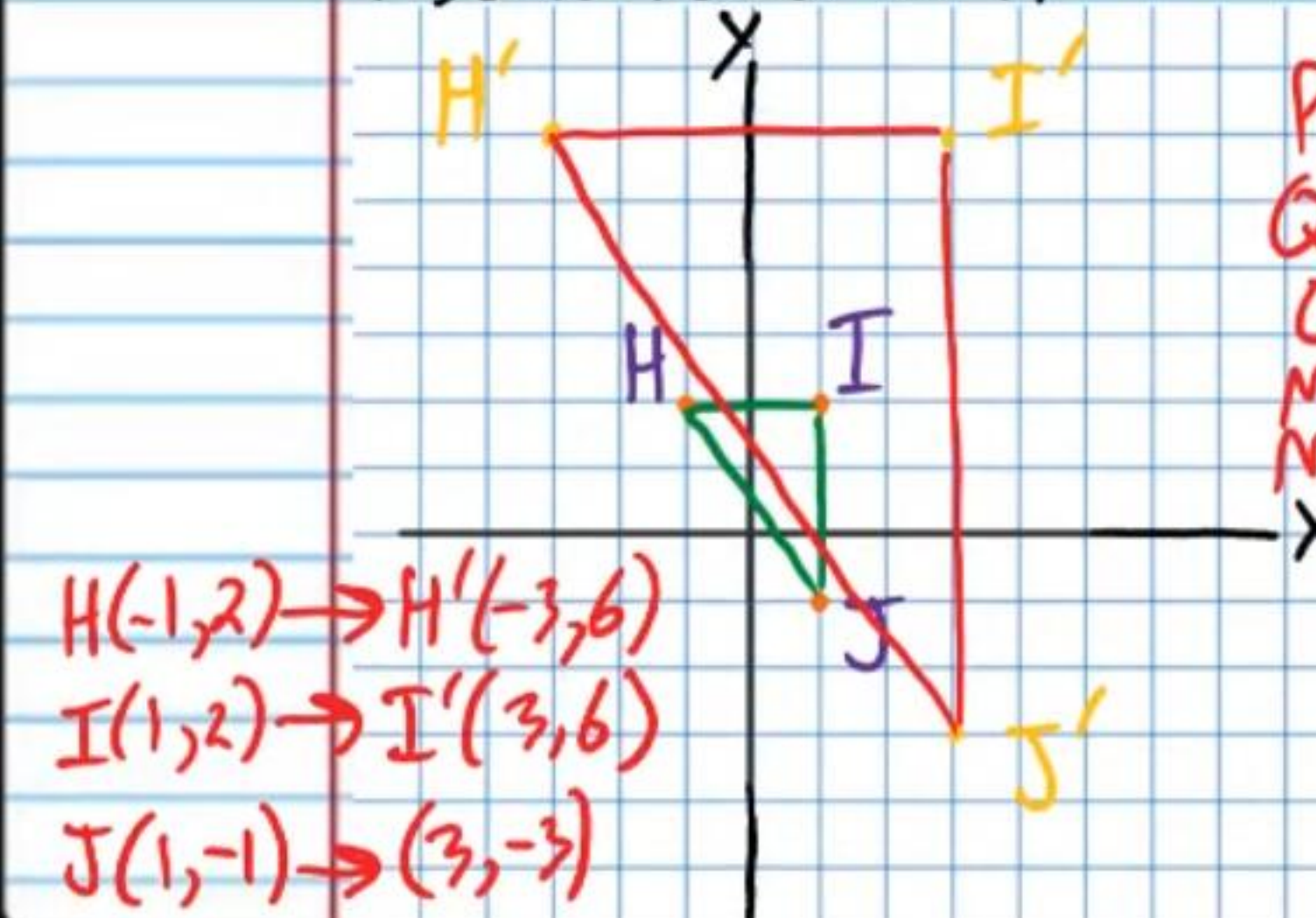
$$\begin{aligned} A(1, 3) &\rightarrow A'(2, 6) \\ B(0, -1) &\rightarrow B'(0, -2) \\ C(3, -1) &\rightarrow C'(6, -2) \end{aligned}$$

⑤⑩ About the origin with a scale factor of $\frac{1}{3}$.



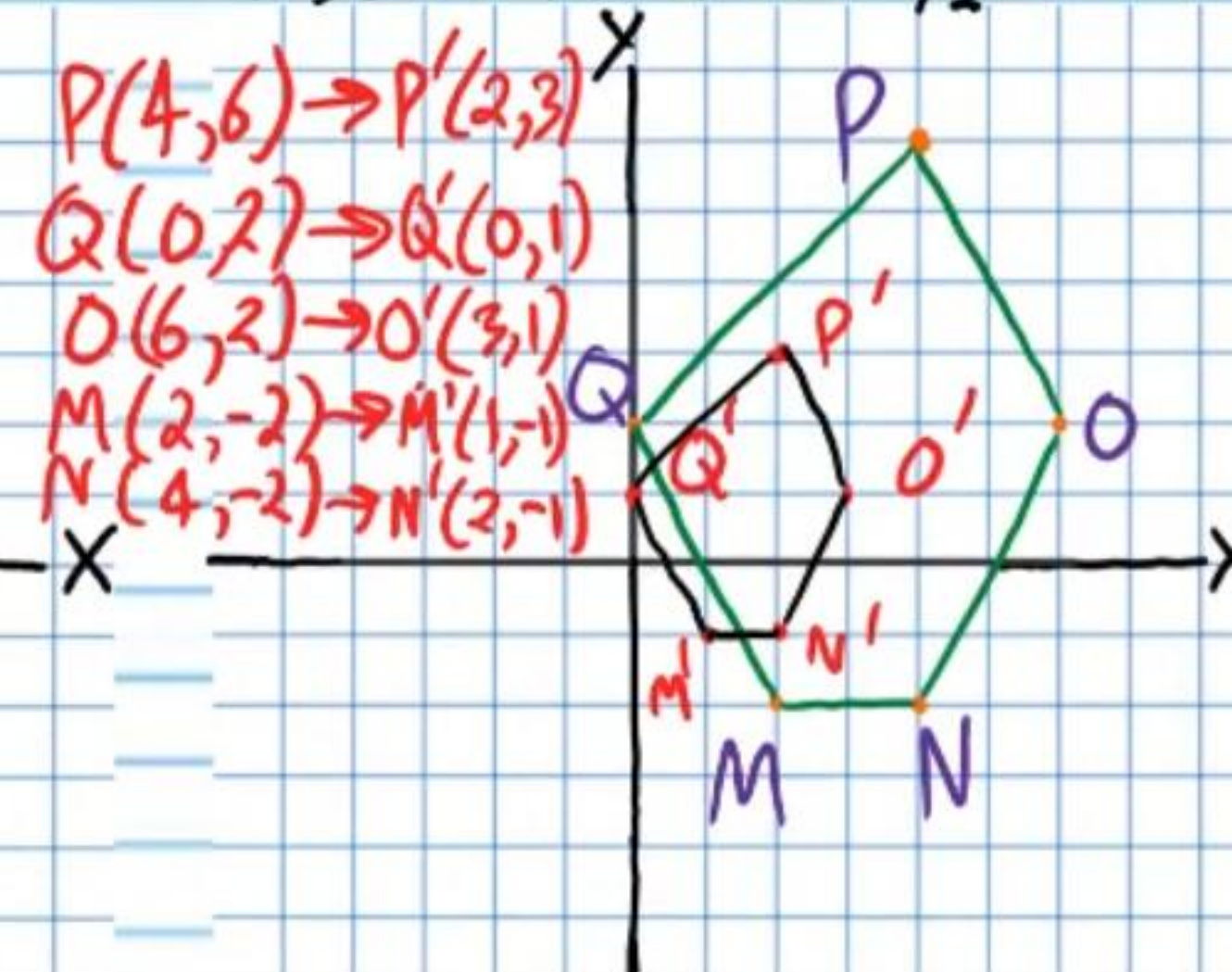
$$\begin{aligned} D(3, 6) &\rightarrow D'(1, 2) \\ F(3, 3) &\rightarrow F'(1, 1) \\ G(6, 3) &\rightarrow G'(2, 1) \\ E(6, 6) &\rightarrow E'(2, 2) \end{aligned}$$

⑤⑪ About the origin with a scale factor of 3.



$$\begin{aligned} H(-1, 2) &\rightarrow H'(-3, 6) \\ I(1, 2) &\rightarrow I'(-3, 6) \\ J(1, -1) &\rightarrow J'(-3, -3) \end{aligned}$$

⑤⑫ About the origin with a scale factor of $\frac{1}{2}$.



$$\begin{aligned} P(4, 6) &\rightarrow P'(2, 3) \\ Q(0, 2) &\rightarrow Q'(0, 1) \\ R(6, 2) &\rightarrow R'(3, 1) \\ S(2, -2) &\rightarrow S'(1, -1) \\ T(4, -2) &\rightarrow T'(2, -1) \end{aligned}$$